

REAL-TIME DATA ANALYTICS • CAPSTONE PROJECT

Smart Energy Analytics Dashboard

A complete, runnable project: ingest smart-meter readings, price them against a time-of-day tariff, learn the building's normal load curve, and detect waste and demand breaches on a live stream of incoming readings — with a recommended action for every alert.

DELIVERABLE	Student handbook + full source code + dataset
LANGUAGE / STACK	Python 3.9+ pandas scikit-learn matplotlib joblib
DATASET	24 hourly baseline readings + 12 live-stream readings (CSV)
RUN COMMAND	<code>pip install -r requirements.txt → cd src → python main.py</code>
LEVEL	Beginner to Intermediate 8-12 hours

Everything needed to build and demonstrate this project is inside this zip file. No internet connection, database or API key required.

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1 Project Overview

Build a system that watches a building's electricity meters in real time, tells the facilities team what every hour is costing them, spots waste the moment it starts, and says exactly what to do about it.

1.1 Problem statement

A commercial building — an office, a campus block, a small factory — pays for electricity through a meter it barely looks at. The bill arrives at the end of the month, it is large, and nobody can say which hour of which day caused it. Three concrete problems follow:

- **Waste is invisible until the bill arrives.** An air conditioner stuck on overnight burns money for weeks before anyone notices a pattern in a monthly total.
- **Nobody knows what an hour costs.** Under a time-of-day tariff the same kilowatt-hour can cost twice as much at 10:00 as at 02:00, so *when* you consume matters as much as *how much*.
- **Demand penalties arrive unannounced.** One machine starting at the wrong moment pushes the building past its contracted demand and adds a penalty that dwarfs the energy charge itself.

This project builds the analytics layer that fixes all three: continuous costing, automatic waste detection, and a live view of demand against the contract.

1.2 Business context

Question the facilities manager asks	How the project answers it
What is this building costing me right now?	Every arriving reading is priced through the time-of-day tariff, the fixed charge and any demand penalty, and the running total updates immediately.
Is this hour normal?	A forecast model predicts what the building should be drawing given the clock, the temperature and the occupancy. The difference is measured in standard deviations of the model's own error.
If it is not normal, what exactly is wrong?	A rule engine names the specific fault — HVAC running in an empty building, machinery left on, demand above contract — in language an engineer can act on.
What should I do about it?	Each finding maps to one recommended action, from dispatching a technician to rescheduling a machinery batch into the off-peak window.
How much am I wasting?	The kWh above forecast on flagged hours is accumulated and converted to rupees at the effective rate.

1.3 What makes this project real-time

A batch report reads a month of data once and prints a summary. A real-time system waits for one reading, reacts to it, updates its state, and waits again. This project reproduces that faithfully:

Real building management system	This project	Where in the code
Meter publishes over MQTT / Modbus every few minutes	<code>live_stream.csv</code> is read row by row with a <code>time.sleep()</code> gap	<code>stream_simulator.run()</code>
Edge gateway derives totals and time features on arrival	<code>engineer_features()</code> runs on each reading	<code>data_loader.py</code>

Real building management system	This project	Where in the code
Billing engine applies the utility's tariff	<code>cost_of_reading()</code> returns energy cost, fixed charge, penalty and carbon	<code>tariff_engine.py</code>
Rule engine raises named alarms	<code>check_rules()</code> returns explicit findings	<code>tariff_engine.py</code>
Deployed model returns an expected load	The persisted pipeline is loaded once and called per reading	<code>model.forecast_one()</code>
SCADA screen refreshes live metrics	<code>RollingKPI</code> recomputes and prints after every reading	<code>stream_simulator.py</code>
Every alarm is written to a historian	Each verdict is appended and saved to <code>realtime_alerts.csv</code>	<code>stream_simulator.run()</code>

Why simulate instead of using a real broker

The event loop is deliberately isolated in one function. Replacing `for _, row in stream.iterrows()` with `for msg in mqtt_client` is the only change needed to make this production-grade — not one line of the costing, rule, model or KPI code would move. That separation is the real lesson.

2 Objectives and Learning Outcomes

2.1 Project objectives

#	Objective
O1	Ingest time-stamped smart-meter data and validate it as a <i>time series</i> : chronological order, constant sampling interval, no gaps.
O2	Engineer the features a load model needs, including a cyclical encoding of the clock so that 23:00 and 00:00 are neighbours.
O3	Implement a time-of-day tariff engine that converts kilowatt-hours into rupees and kilograms of carbon dioxide.
O4	Implement explainable operating rules that name specific waste conditions rather than merely flagging a number.
O5	Explore the day's load curve statistically and visually to find where the energy and the money actually go.
O6	Train and cross-validate a load forecasting model, and derive an anomaly threshold from its own residual spread.
O7	Process a live stream of readings, classify each as NORMAL, WATCH or ALERT, recommend an action, and maintain rolling KPIs.
O8	Produce an auditable output set: charts, a saved model, an alert log and a summary report.

2.2 What you will learn

Area	Skills demonstrated
Time-series data handling	<code>parse_dates</code> , sorting chronologically, detecting the sampling interval with <code>diff()</code> , filling gaps by interpolation rather than by a column median
Feature engineering for time	Extracting hour and day from a timestamp, flagging working hours, and the sine/cosine cyclical encoding that makes the clock circular
Domain costing	Time-of-day tariff slabs, fixed charges, contracted demand and penalty arithmetic, grid emission factors, load factor
Machine learning	<code>Pipeline</code> , <code>StandardScaler</code> , <code>LinearRegression</code> , K-fold cross-validation with <code>cross_val_predict</code> , MAE / RMSE / MAPE / R-squared, coefficient interpretation, model persistence
Anomaly detection	Residual standard deviation as an adaptive threshold, sigma scoring, and why that beats a fixed kWh cut-off
Visualisation	Stacked area load curve, conditionally coloured bar chart, scatter with a colour-mapped third variable, horizontal ranked bars
Real-time systems thinking	Event loops, per-reading scoring, O(1) rolling aggregates, arbitration between a rule engine and a model, alert logging
Critical analysis	Spotting multicollinearity when a coefficient contradicts a correlation, and knowing what a single day of history can and cannot support

3 Project Scope

In scope	Out of scope
CSV-based ingestion of hourly sub-meter readings	Live MQTT / Modbus / BACnet connection to physical meters
Time-of-day tariff costing with fixed charge and demand penalty	Full utility billing including taxes, duties and power-factor rebates
Carbon accounting from a fixed grid emission factor	Certified emissions reporting or renewable-energy certificates
Rule-based detection of named waste conditions	Non-intrusive load disaggregation down to individual appliances
Hourly load forecasting from clock, temperature and occupancy	Multi-day-ahead forecasting or weather-forecast integration
Residual-based anomaly scoring in sigma units	Root-cause diagnosis or automated equipment control
Simulated real-time stream driven by a CSV and a timer	A production message broker and a historian database
Rolling KPIs printed to the console and written to CSV	A hosted multi-user SCADA dashboard with paging and escalation
An optional Streamlit dashboard	Authentication, role-based access or deployment

Deliberately small dataset

One 24-hour baseline day is far below what a real energy model needs (a year, ideally three, to capture seasons, weekends and holidays). It is intentional: you can read every reading, verify every rupee by hand, and still exercise the complete pipeline.

Section 13 states precisely which conclusions this supports and which it does not, and Section 17 shows how to scale it up.

4 Requirement Specification

This is the contract the code must satisfy. Every requirement is implemented, and the module that implements it is named so you can verify it yourself.

4.1 Functional requirements

ID	Requirement	Implemented in
FR-01	The system shall load timestamped meter readings from CSV, parsing the timestamp column as a datetime, and abort with a clear message if the file is missing.	<code>data_loader.load_csv</code>
FR-02	The system shall reject impossible readings: a negative meter value shall be treated as a sensor fault.	<code>data_loader.validate</code>
FR-03	The system shall detect duplicate timestamps and keep only the first reading for each.	<code>data_loader.validate</code>
FR-04	The system shall fill missing readings by interpolating between neighbouring hours, not by a column average.	<code>data_loader.validate</code>
FR-05	The system shall sort readings chronologically and report the detected sampling interval and any gaps in the series.	<code>data_loader.validate</code>
FR-06	The system shall compute the building total from the four sub-meters and derive hour, working-hour flag and cyclical hour features.	<code>data_loader.engineer_features</code>
FR-07	The system shall assign each reading its time-of-day tariff slab and rate.	<code>tariff_engine.slabs_for_hour</code>
FR-08	The system shall compute the hourly bill as energy charge + fixed charge + demand penalty, and the carbon emitted.	<code>tariff_engine.cost_of_reading</code>
FR-09	The system shall apply operating rules that detect HVAC, machinery and lighting waste in an unoccupied building, demand above contract, and peak-window machinery use.	<code>tariff_engine.check_rules</code>
FR-10	The system shall map findings to exactly one recommended action per reading.	<code>tariff_engine.recommend</code>
FR-11	The system shall report total, average, peak and base load, load factor, consumption per sub-meter and cost per tariff slab.	<code>eda.print_summary</code>
FR-12	The system shall generate and save at least four analytical charts as PNG files.	<code>eda.build_charts</code>
FR-13	The system shall train a load forecasting model inside a scikit-learn Pipeline so scaling is identical at training and prediction time.	<code>model.build_pipeline</code>
FR-14	The system shall evaluate the model by K-fold cross-validation and report MAE, RMSE, MAPE and R-squared.	<code>model.train</code>
FR-15	The system shall derive the anomaly threshold from the standard deviation of the model's own out-of-fold residuals.	<code>model.train</code>
FR-16	The system shall persist the fitted pipeline together with its error band and reload both without retraining.	<code>model.train / model.load</code>
FR-17	The system shall classify each reading's deviation as NORMAL, WATCH or ALERT in sigma units.	<code>model.classify_deviation</code>
FR-18	The system shall process incoming readings one at a time with a configurable delay between them.	<code>stream_simulator.run</code>
FR-19	The system shall combine the rule findings and the model deviation into one status, with a named rule finding able to raise severity on its own.	<code>stream_simulator.combine_status</code>
FR-20		<code>stream_simulator.RollingKPI</code>

ID	Requirement	Implemented in
	The system shall maintain rolling KPIs — energy, cost, carbon, penalty, peak demand, alert count and avoidable waste — updated after every reading.	
FR-21	The system shall write every reading's verdict, findings and recommended action to an audit CSV.	<code>stream_simulator.run</code>
FR-22	The system shall write a plain-text summary report of the whole run.	<code>main.write_summary</code>
FR-23	The system shall expose command-line flags to skip the delay and to skip chart generation.	<code>main.main</code>

4.2 Non-functional requirements

ID	Category	Requirement
NFR-01	Explainability	No alert may be raised without either a named rule finding or a quantified deviation, plus a recommended action.
NFR-02	Configurability	Every tariff rate, threshold and path lives in <code>config.py</code> . No magic numbers anywhere else.
NFR-03	Modularity	Each module runs standalone via its <code>__main__</code> block for isolated testing.
NFR-04	Reproducibility	A fixed <code>random_state</code> on the cross-validation folds makes every run produce identical numbers.
NFR-05	Adaptivity	The anomaly threshold is derived from the model's own error, so it tightens automatically as the model improves.
NFR-06	Portability	Runs on Windows, macOS, Linux and Colab with no code changes; matplotlib uses the headless Agg backend.
NFR-07	Performance	End-to-end run finishes in under 5 seconds with <code>--fast</code> ; per-reading scoring is a few milliseconds and each KPI update is $O(1)$.
NFR-08	Auditability	Every run leaves charts, a model file, an alert log and a summary report on disk.
NFR-09	Robustness	Missing files raise a descriptive error; negative and missing readings are repaired before they reach the model.

4.3 Software and hardware requirements

Item	Requirement
Operating system	Windows 10+, macOS 11+, or any modern Linux
Python	3.9 or newer (3.10 / 3.11 recommended)
RAM	2 GB is more than sufficient
Disk	About 60 MB including the Python libraries
Internet	Needed once, only to <code>pip install</code> the libraries. The project itself runs fully offline.
Editor	VS Code, PyCharm, Jupyter or plain IDLE
Libraries	pandas, numpy, scikit-learn, matplotlib, joblib (streamlit only for the optional dashboard)

REQUIREMENTS.TXT

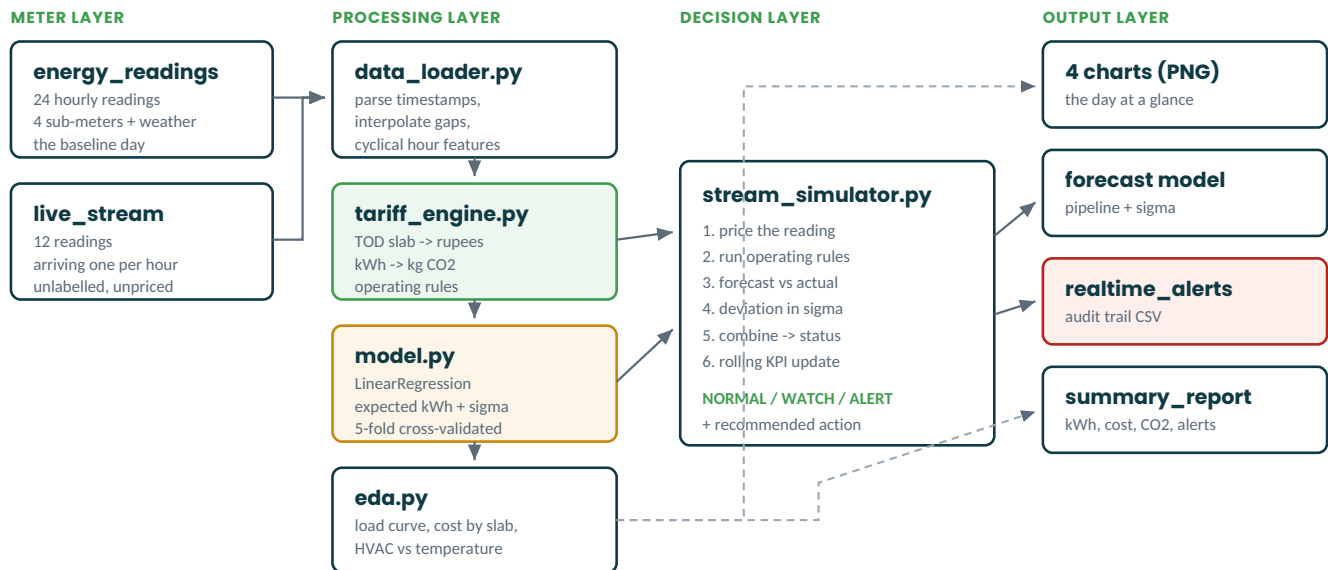
```
pandas>=2.0
numpy>=1.24
scikit-learn>=1.3
```

```
matplotlib>=3.7
joblib>=1.3

# optional, only for app/dashboard.py
# streamlit>=1.30
```

5 System Architecture

Four layers, data flowing strictly left to right. No layer reaches backwards, which is what allows any single stage to be tested, replaced or scaled without touching the others.



Layer	Responsibility	Modules
Meter	Provide raw timestamped readings, historical and live	data/*.csv
Processing	Clean and enrich the series, price it, learn the normal load curve, produce analytics	data_loader.py, tariff_engine.py, model.py, eda.py
Decision	Arbitrate between the rules and the model, issue a status and an action, track live KPIs	stream_simulator.py
Output	Persist everything an auditor or an energy manager would ask for	outputs/

The two-brain design

The rule engine and the forecast model are deliberately independent. The rule engine is *deterministic and specific* — it can say 'the HVAC is running in an empty building' — but it only catches faults somebody thought of in advance. The model is *general* — it flags any hour that does not look like the others — but it cannot say why. Running both and combining their verdicts gives you named faults where rules exist and a safety net where they do not.

6 Dataset Description

Two CSV files ship with the project. Both use the same schema. The baseline file is the day the system learns from; the stream file is the day it has to judge.

File	Rows	Period	Used for
data/energy_readings.csv	24	2026-03-09, 00:00 to 23:00	Tariff calibration, EDA, model training and cross-validation
data/live_stream.csv	12	2026-03-10, 00:00 to 11:00	The simulated live feed of arriving readings

6.1 Data dictionary

Column	Type	Unit	Description
timestamp	datetime	—	Start of the hour the reading covers
hvac_kwh	float	kWh	Sub-meter 1: air conditioning, ventilation and heating
lighting_kwh	float	kWh	Sub-meter 2: all lighting circuits
machinery_kwh	float	kWh	Sub-meter 3: production or process equipment
plug_load_kwh	float	kWh	Sub-meter 4: computers, printers and everything on a socket
temperature_c	float	degrees C	Outside air temperature, the main driver of HVAC load
occupancy	int	people	People counted in the building during that hour

Why four sub-meters instead of one building total

A single total tells you the bill is high. Four sub-meters tell you *which system* is responsible, which is the difference between a report and an action. Splitting the meter is the single highest-value change most buildings can make to their monitoring.

6.2 Sample readings

DATA/ENERGY_READINGS.CSV (EXTRACT)

```
timestamp,hvac_kwh,lighting_kwh,machinery_kwh,plug_load_kwh,temperature_c,occupancy
2026-03-09 00:00,2.1,0.8,0.5,0.6,24.5,0
2026-03-09 01:00,2.0,0.8,0.5,0.6,24.0,0
2026-03-09 02:00,1.9,0.8,0.5,0.6,23.6,0
2026-03-09 03:00,1.9,0.8,0.5,0.6,23.2,0
2026-03-09 04:00,2.0,0.8,0.5,0.6,23.0,0
2026-03-09 05:00,2.2,1.0,0.6,0.7,23.4,2
2026-03-09 06:00,2.6,1.6,1.2,1.1,24.2,8
2026-03-09 07:00,3.4,2.8,3.5,2.0,25.6,22
```

6.3 Engineered features

Feature	Formula	Why it exists
total_kwh	sum of the four sub-meters	The building's demand for that hour; the target the model predicts and the number the tariff is applied to.
hour	timestamp.dt.hour	Needed to look up the time-of-day tariff slab.
is_working_hour	1 if 08:00 ≤ hour ≤ 18:00	A step change that no smooth function of the clock can capture: the building simply switches on.

Feature	Formula	Why it exists
hour_sin	$\sin(2\pi \cdot \text{hour} / 24)$	First half of the cyclical clock encoding — see below.
hour_cos	$\cos(2\pi \cdot \text{hour} / 24)$	Second half. Together the pair places every hour on a circle.
*_share	sub-meter kWh / total_kwh	Each system's share of the hour, used by the charts.

6.4 The cyclical time trick (the key idea in this project)

Feed `hour` to a model as a plain number from 0 to 23 and you have told it that 23:00 and 00:00 are **23 units apart**, when in reality they are one hour apart. The model then learns a straight line across a boundary that does not exist.

The fix is to place each hour on a circle and hand over its coordinates:

DATA_LOADER.ENGINEER_FEATURES

```
angle = 2 * np.pi * df["hour"] / 24
df["hour_sin"] = np.sin(angle)
df["hour_cos"] = np.cos(angle)
```

Hour	Raw value	hour_sin	hour_cos	Distance to hour 0
0	0	0.000	1.000	0.00
6	6	1.000	0.000	1.41
12	12	0.000	-1.000	2.00 (opposite side)
18	18	-1.000	0.000	1.41
23	23	-0.259	0.966	0.26 — correctly close

Read the last row carefully

As a raw number, hour 23 is the furthest possible point from hour 0. In sine/cosine space it is the *nearest*. That one change is why the model in Section 13 reaches an R-squared of 0.997 on a load curve that wraps around midnight. The same technique applies to day of week, month of year and compass bearing.

7 Project Folder Structure

FOLDER TREE

```

SmartEnergyAnalytics/
|
+-- README.md                quick start guide
+-- requirements.txt          library versions
+-- Smart_Energy_Analytics_Dashboard.pdf  this handbook
|
+-- data/
|   +-- energy_readings.csv    24 hourly readings = one baseline day
|   +-- live_stream.csv        12 incoming readings for the live feed
|
+-- src/
|   +-- config.py              paths, tariff slabs, thresholds, feature lists
|   +-- data_loader.py         ingestion, validation, time-series features
|   +-- tariff_engine.py        cost, carbon and explainable operating rules
|   +-- eda.py                 console statistics + 4 matplotlib charts
|   +-- model.py               load forecast + cross-validated error band
|   +-- stream_simulator.py     real-time loop with rolling KPIs
|   +-- main.py                orchestrator <-- RUN THIS
|
+-- app/
|   +-- dashboard.py           OPTIONAL Streamlit control room
|
+-- sample_outputs/           reference results, for comparison only
|   +-- console_output.txt     the full console log of a reference run
|   +-- ... the same 7 files listed below
|
+-- outputs/                   EMPTY until you run the project
    +-- 01_load_curve.png
    +-- 02_cost_by_slab.png
    +-- 03_hvac_vs_temperature.png
    +-- 04_consumption_share.png
    +-- load_forecast_model.joblib
    +-- realtime_alerts.csv
    +-- summary_report.txt
  
```

Why src/ has no package prefix

The modules import each other plainly (`import config`) and you run `python main.py` from inside `src/` . This keeps imports readable for beginners. In a production repository you would add an `__init__.py` , turn `src` into a package and import as `from src import config` .

8 Methodology: The Five Stages

Running `python main.py` executes these five stages in order, each printed with a banner so you can follow along.

Stage	Name	Module	What happens
Stage 1	Ingestion and time features	<code>data_loader.py</code>	Parse timestamps, repair negative and missing readings by interpolation, sort chronologically, confirm the hourly interval, then derive the total, the hour, the working-hour flag and the cyclical sine/cosine pair.
Stage 2	Tariff costing and rules	<code>tariff_engine.py</code>	Look up each hour's tariff slab, compute energy charge, fixed charge, demand penalty, total cost and carbon; run the operating rules that name specific waste conditions.
Stage 3	Exploratory analysis	<code>eda.py</code>	Print totals, peak and base load, load factor, consumption per sub-meter and cost per slab; save four charts. This is where you form hypotheses before letting a model near the data.
Stage 4	Load forecasting	<code>model.py</code>	Scale the features, cross-validate a linear model over five folds, report MAE / RMSE / MAPE / R-squared, take the residual standard deviation as the anomaly yardstick, refit on the whole day and persist the bundle.
Stage 5	Real-time stream	<code>stream_simulator.py</code>	Consume readings one at a time: price, rule-check, forecast, score the deviation in sigma units, arbitrate into a status, recommend an action, update rolling KPIs, print a live block, append to the alert log.

8.1 Data flow in one sentence per hop

1. CSV file → `pandas.DataFrame` with a parsed `timestamp` column (7 columns).
2. DataFrame → validated series (chronological, no duplicates, no negatives, gaps interpolated).
3. Validated series → enriched frame (total, hour, working-hour flag, sine/cosine, per-meter shares).
4. Enriched frame → priced frame (slab, rate, energy cost, penalty, total cost, carbon).
5. Priced frame → feature matrix X and target y → cross-validated pipeline + residual sigma → `load_forecast_model.joblib`.
6. Stream reading → enriched Series → cost + rule findings + expected kWh → sigma score → status and action → KPI update → log row.

9 The Tariff and Operating Rulebook

Every rate and threshold below lives in `config.py`. Change the bank's appetite — sorry, the building's tariff — by editing one file and re-running.

9.1 Time-of-day tariff slabs

Utilities charge more when the grid is stressed. Two peaks a day is the normal Indian TOD structure: a morning industrial peak and an evening domestic one.

Slab	Hours covered	Rate	Why
PEAK	09:00–11:59 and 18:00–20:59	Rs. 10.80 / kWh	Grid demand is highest; the utility is running its most expensive generation
NORMAL	06:00–08:59, 12:00–17:59, 21:00–21:59	Rs. 7.50 / kWh	Ordinary daytime load
OFF-PEAK	22:00–05:59	Rs. 5.20 / kWh	Overnight; the utility wants to sell surplus

9.2 How an hourly bill is built

Component	Formula	Value in this project
Energy charge	kWh × slab rate	Varies by hour
Fixed charge	A flat standing charge every hour	Rs. 12.00
Demand penalty	$\max(0, \text{kWh} - \text{contracted demand}) \times \text{penalty rate}$	Contract 25 kWh, penalty Rs. 45.00 per excess kWh
Hourly bill	energy + fixed + penalty	
Carbon	kWh × grid emission factor	0.82 kg CO ₂ per kWh

Why the demand penalty matters so much

At 31.8 kWh in a single hour the building is 6.8 kWh over its 25 kWh contract. At Rs.45 per excess kWh that is Rs.306 of penalty on top of Rs.343 of energy — the penalty is almost as large as the electricity itself. This is why demand management, not just consumption reduction, is where the money is.

9.3 Operating rules

These are the conditions a facilities engineer would recognise instantly. Each returns a sentence, not a score.

Rule	Trigger condition	What it means
HVAC in an empty building	occupancy ≤ 2 and hvac_kwh > 3.0	Stuck damper, failed thermostat, or a schedule nobody updated
Machinery left running	occupancy ≤ 2 and machinery_kwh > 1.5	Equipment not shut down at the end of the shift
Lights left on	occupancy ≤ 2 and lighting_kwh > 1.5	No occupancy sensor, or one that is not working
Demand breach	total_kwh > 25.0	The contracted demand is exceeded and a penalty is being charged
Peak-window machinery	slab is PEAK and machinery_kwh > 12.0	A batch that could run overnight is running at the most expensive rate of the day

9.4 The anomaly band

The rules only catch faults somebody anticipated. The model catches the rest, by comparing what actually happened against what it expected:

Deviation	Status	Meaning
Within 2σ	NORMAL	Indistinguishable from ordinary variation
2σ to 3σ	WATCH	Unusual; worth a look but not yet actionable
Beyond 3σ	ALERT	Statistically almost impossible under normal operation

Why sigma instead of a fixed kWh threshold

σ is the standard deviation of the model's own prediction error, measured on out-of-fold data. In this run $\sigma = 0.497$ kWh, so ALERT fires beyond a 1.49 kWh error. If you later improve the model, σ shrinks and the detector automatically becomes more sensitive — with no threshold to re-tune by hand. A hard-coded '2 kWh' would silently become too loose or too tight the moment anything changed.

9.5 Arbitration: how the two verdicts combine

Rule findings	Model status	Final status	Reasoning
Yes	ALERT	ALERT	Both layers agree — the strongest possible evidence
Yes	NORMAL or WATCH	ALERT	A named fault is hard evidence; it does not need the model's permission
No	ALERT	WATCH	Deliberately downgraded: a purely statistical outlier with no known cause, on one day of history, is a hint rather than a fault
No	WATCH	WATCH	Mild deviation
No	NORMAL	NORMAL	Within the expected band

9.6 A worked example: the 03:00 reading

On 10 March at 03:00 the meters report HVAC 6.9, lighting 0.9, machinery 0.6, plug 0.7 kWh, at 23.2°C with nobody in the building.

Step	Calculation	Result
Total demand	$6.9 + 0.9 + 0.6 + 0.7$	9.1 kWh
Tariff slab	hour 3 falls in 22:00–05:59	OFF-PEAK at Rs. 5.20
Energy charge	9.1×5.20	Rs. 47.32
Fixed charge	flat	Rs. 12.00
Demand penalty	9.1 is below the 25 kWh contract	Rs. 0.00
Hourly bill	$47.32 + 12.00$	Rs. 59.32
Model expectation	clock + 23.2°C + 0 people	3.6 kWh
Deviation	$9.1 - 3.6$	+5.5 kWh, or +151%
Sigma score	$5.5 / 0.497$	+11.0σ
Rule finding	occupancy 0 and HVAC 6.9 > 3.0	HVAC fault suspected

Step	Calculation	Result
Verdict	rule finding + model ALERT	ALERT
Action	mapped from the finding	Dispatch technician to inspect the HVAC unit now

This is what a useful alert looks like

Not 'anomaly detected, score 0.87'. Instead: *at 03:00 the HVAC drew 6.9 kWh with nobody in the building; that is 151% above expectation and 11 standard deviations out; it cost Rs.59.32 this hour; send a technician.* Every number in that sentence comes from a different module, and each one is independently checkable.

10 Code Walkthrough

Seven modules, about 750 lines in total. Each is printed here in full and then explained. Read them in this order — each builds on the one before it.

Order	Module	Lines	One-line job
1	config.py	~95	Every path, tariff rate and threshold in one place
2	data_loader.py	~110	Read, validate and enrich the meter series
3	tariff_engine.py	~120	kWh → rupees, carbon and named waste findings
4	eda.py	~135	Describe the day in numbers and four charts
5	model.py	~130	Forecast the load and derive the anomaly band
6	stream_simulator.py	~150	The real-time decision loop
7	main.py	~110	Wire the five stages together

10.1 config.py — the single source of truth

An energy manager will want to argue about the tariff rates, the contracted demand and the idle thresholds. None of those arguments should require reading the analytics code. **If it is a number somebody could dispute, it lives here.**

SRC/CONFIG.PY

```
"""
config.py
-----
One place for every path, tariff rate, threshold and column list used by the
project. If a number could be argued about by an energy manager, it lives here
and nowhere else.
"""

from pathlib import Path

# -----
# 1. FOLDER LAYOUT
# -----
BASE_DIR = Path(__file__).resolve().parent.parent
DATA_DIR = BASE_DIR / "data"
OUTPUT_DIR = BASE_DIR / "outputs"

BASELINE_CSV = DATA_DIR / "energy_readings.csv"
STREAM_CSV = DATA_DIR / "live_stream.csv"

MODEL_FILE = OUTPUT_DIR / "load_forecast_model.joblib"
ALERT_LOG = OUTPUT_DIR / "realtime_alerts.csv"
SUMMARY_REPORT = OUTPUT_DIR / "summary_report.txt"

OUTPUT_DIR.mkdir(parents=True, exist_ok=True)

# -----
# 2. COLUMN GROUPS
# -----
# The four sub-meters installed in the building
LOAD_COLUMNS = ["hvac_kwh", "lighting_kwh", "machinery_kwh", "plug_load_kwh"]

# Friendly labels for charts and console output
LOAD_LABELS = {
    "hvac_kwh": "HVAC",
```

```

    "lighting_kwh": "Lighting",
    "machinery_kwh": "Machinery",
    "plug_load_kwh": "Plug load",
}

TARGET = "total_kwh"

# Inputs given to the forecasting model
FEATURES = [
    "hour_sin",          # cyclical encoding of the clock
    "hour_cos",
    "temperature_c",
    "occupancy",
    "is_working_hour",
]

# -----
# 3. TARIFF: TIME-OF-DAY SLABS (Indian TOD structure, rupees per kWh)
# -----
# Each entry: (slab name, set of hours it covers, rate per kWh)
TARIFF_SLABS = [
    ("PEAK",      {9, 10, 11, 18, 19, 20},          10.80),
    ("NORMAL",    {6, 7, 8, 12, 13, 14, 15, 16, 17, 21}, 7.50),
    ("OFF-PEAK",  {22, 23, 0, 1, 2, 3, 4, 5},        5.20),
]
DEFAULT_RATE = 7.50

FIXED_CHARGE_PER_HOUR = 12.0      # standing charge billed every hour
CO2_KG_PER_KWH = 0.82             # Indian grid emission factor

# -----
# 4. OPERATING RULES
# -----
CONTRACTED_DEMAND_KWH = 25.0      # exceeding this in one hour draws a penalty
DEMAND_PENALTY_PER_KWH = 45.0     # rupees for every kWh above the contract

WORKING_HOURS = range(8, 19)      # 08:00 to 18:00 inclusive of 18

# Rule-engine thresholds (used by tariff_engine.check_rules)
IDLE_OCCUPANCY = 2                # at or below this, the building is "empty"
HVAC_IDLE_LIMIT_KWH = 3.0         # HVAC above this with nobody in = fault
MACHINERY_IDLE_LIMIT_KWH = 1.5    # machinery running out of hours
LIGHTING_IDLE_LIMIT_KWH = 1.5    # lights left on

# -----
# 5. MODEL AND ANOMALY SETTINGS
# -----
CV_FOLDS = 5
RANDOM_STATE = 42

# Deviation measured in standard deviations of the model's own error
WATCH_SIGMA = 2.0                 # 2 sigma -> keep an eye on it
ALERT_SIGMA = 3.0                 # 3 sigma -> raise an alert

STREAM_DELAY_SECONDS = 1.0        # pause between two simulated readings

```

Construct	What it does and why it matters
<pre>BASE_DIR = Path(__file__).resolve().parent.parent</pre>	Builds every other path relative to the project root rather than the current working directory, so the project runs identically from <code>SRC/</code> , from the root, or from an IDE.
TARIFF_SLABS	The tariff expressed as <i>data</i> — a list of (name, set of hours, rate) tuples — rather than as an <code>if/elif</code> chain. Changing the utility's rate card never means changing logic. Using a <code>set</code> of hours also makes the lookup $O(1)$.

Construct	What it does and why it matters
LOAD_COLUMNS and LOAD_LABELS	The four sub-meters as a list, plus display names. Adding a fifth meter is a one-line change here; the totals, the charts and the shares all follow automatically.
FEATURES	Exactly what the model is allowed to see. Note that the raw hour is not in the list — only its sine and cosine, for the reason given in Section 6.4.
WATCH_SIGMA / ALERT_SIGMA	Thresholds expressed in standard deviations, not kilowatt-hours, so they stay meaningful when the model changes.
OUTPUT_DIR.mkdir(exist_ok=True)	Runs at import time, so no other module has to check whether outputs/ exists.

10.2 data_loader.py — ingestion and time-series features

Time-series data brings two jobs ordinary tabular data does not: the timestamp must be parsed and trusted, and the clock has to be turned into something a model can learn from.

SRC/DATA_LOADER.PY

```

"""
data_loader.py
-----
Stage 1: read the meter file, prove it is usable, and derive the columns the
tariff engine and the forecast model need.

Time-series data brings two jobs that ordinary tabular data does not:
parsing the timestamp properly, and turning the clock into something a model
can actually learn from.
"""

import numpy as np
import pandas as pd

import config

def load_csv(path):
    """Read a meter CSV and fail loudly if it is missing."""
    if not path.exists():
        raise FileNotFoundError(f"Required data file not found: {path}")
    df = pd.read_csv(path, parse_dates=["timestamp"])
    print(f"[data_loader] Loaded {len(df)} readings from {path.name} "
          f"from {df['timestamp'].min()} to {df['timestamp'].max()}")
    return df

def validate(df):
    """Data-quality gate. Returns the cleaned frame plus a small report."""
    report = {
        "readings": len(df),
        "duplicate_timestamps": int(df["timestamp"].duplicated().sum()),
        "missing_values": int(df.isna().sum().sum()),
        "negative_readings": int((df[config.LOAD_COLUMNS] < 0).sum().sum()),
    }

    # A meter can never run backwards; a negative value is a sensor fault.
    for col in config.LOAD_COLUMNS:
        df.loc[df[col] < 0, col] = np.nan

    # Duplicate timestamps mean the meter reported twice: keep the first.
    if report["duplicate_timestamps"] > 0:
        df = df.drop_duplicates(subset="timestamp", keep="first")

    # A dropped reading is filled by interpolating between its neighbours,

```

```

# which suits time series far better than a column median.
numeric = config.LOAD_COLUMNS + ["temperature_c", "occupancy"]
before = int(df[numeric].isna().sum().sum())
if before:
    df[numeric] = df[numeric].interpolate(limit_direction="both")
report["interpolated"] = before

# Readings must be in chronological order for any time-based logic.
df = df.sort_values("timestamp").reset_index(drop=True)

# Sampling interval check: are the readings really hourly?
gaps = df["timestamp"].diff().dropna().dt.total_seconds() / 3600
report["interval_hours"] = float(gaps.mode().iloc[0]) if len(gaps) else 0.0
report["gaps_in_series"] = int((gaps != report["interval_hours"]).sum())

print(f"[data_loader] Validation -> {report}")
return df, report

def engineer_features(df):
    """Add the derived columns used everywhere downstream."""
    df = df.copy()

    # Total draw of the building for that hour: the headline number.
    df["total_kwh"] = df[config.LOAD_COLUMNS].sum(axis=1).round(2)

    # Calendar parts pulled out of the timestamp.
    df["hour"] = df["timestamp"].dt.hour
    df["day"] = df["timestamp"].dt.date
    df["is_working_hour"] = df["hour"].isin(config.WORKING_HOURS).astype(int)

    # Cyclical encoding. Hour 23 and hour 0 are one hour apart, but as plain
    # numbers they look 23 apart. Mapping the clock onto a circle with sine and
    # cosine tells the model that midnight follows 23:00.
    angle = 2 * np.pi * df["hour"] / 24
    df["hour_sin"] = np.sin(angle).round(4)
    df["hour_cos"] = np.cos(angle).round(4)

    # Share of the total taken by each sub-meter, useful for the charts.
    for col in config.LOAD_COLUMNS:
        df[col.replace("_kwh", "_share")] = (df[col] / df["total_kwh"]).round(3)

    print(f"[data_loader] Engineered features; total load "
          f"{df['total_kwh'].sum():.1f} kWh over {len(df)} hours")
    return df

def get_baseline():
    """Full pipeline for the historical day used to learn normal behaviour."""
    df, _ = validate(load_csv(config.BASELINE_CSV))
    return engineer_features(df)

def get_stream():
    """Same treatment for the live feed, so the model sees identical inputs."""
    df, _ = validate(load_csv(config.STREAM_CSV))
    return engineer_features(df)

if __name__ == "__main__":
    data = get_baseline()
    print(data[["timestamp", "total_kwh", "hour_sin", "hour_cos",
               "is_working_hour"]].head(8).to_string(index=False))

```

Element	Explanation
<code>pd.read_csv(path, parse_dates=["timestamp"])</code>	Converts the timestamp column to real datetimes at load time. Without this it stays a string and <code>.dt.hour</code> fails.
Negative-reading check	A meter cannot run backwards. Any negative value is set to <code>NaN</code> so it is repaired by the interpolation step rather than silently poisoning the totals.
<code>.interpolate(limit_direction="both")</code>	The key difference from the tabular case. A missing 14:00 reading is best estimated from 13:00 and 15:00, not from the day's median. Interpolation respects the shape of the load curve; a median would flatten it.
<code>sort_values("timestamp")</code>	Every downstream step assumes chronological order. Sorting once here means nothing later has to worry about it.
Sampling-interval check	<code>timestamp.diff()</code> gives the gap between consecutive readings; the mode is the true interval and anything else is a gap in the series. Real meter feeds drop readings constantly, and a silent gap will quietly corrupt any time-based aggregate.
<code>total_kwh = df[LOAD_COLUMNS].sum(axis=1)</code>	<code>axis=1</code> sums across the four meter columns for each row, giving the building total for that hour.
<code>is_working_hour</code>	A binary step. The building switching on at 08:00 is a discontinuity that no smooth function of the clock can represent, so it gets its own feature.
<code>hour_sin/hour_cos</code>	The cyclical encoding from Section 6.4. Two columns replace one, and midnight stops being a cliff edge.
<code>get_baseline()/get_stream()</code>	Thin wrappers chaining <code>load</code> → <code>validate</code> → <code>engineer</code> . Routing both files through the same functions is the cheapest possible insurance against training/serving skew .

10.3 tariff_engine.py – money, carbon and named faults

No machine learning at all: deterministic arithmetic an electricity bill auditor could check by hand. That is precisely why it sits alongside the model rather than inside it.

SRC/TARIFF_ENGINE.PY

```
"""
tariff_engine.py
-----
Stage 2: turn kilowatt-hours into money, carbon and plain-English warnings.

This module contains no machine learning at all. It is deterministic
arithmetic that an electricity bill auditor could check by hand, which is
exactly why it sits alongside the model rather than inside it.
"""

import config

def slab_for_hour(hour):
    """Return (slab name, rupees per kWh) for a given clock hour."""
    for name, hours, rate in config.TARIFF_SLABS:
        if hour in hours:
            return name, rate
    return "NORMAL", config.DEFAULT_RATE

def cost_of_reading(row):
    """
    Price one hourly reading.

    Bill for the hour = energy charge + fixed charge + demand penalty
    """
```

```

    energy_charge = kWh consumed x the slab rate for that hour
    fixed_charge = a flat standing charge
    demand_penalty = charged only on the kWh above the contracted demand
    """
    slab, rate = slab_for_hour(int(row["hour"]))
    kWh = float(row["total_kwh"])

    energy_cost = kWh * rate
    excess = max(0.0, kWh - config.CONTRACTED_DEMAND_KWH)
    penalty = excess * config.DEMAND_PENALTY_PER_KWH

    return {
        "tariff_slab": slab,
        "tariff_rate": rate,
        "energy_cost": round(energy_cost, 2),
        "demand_excess_kwh": round(excess, 2),
        "demand_penalty": round(penalty, 2),
        "total_cost": round(energy_cost + config.FIXED_CHARGE_PER_HOUR + penalty, 2),
        "co2_kg": round(kWh * config.CO2_KG_PER_KWH, 2),
    }

def check_rules(row):
    """
    Explainable operating rules. Each one describes a specific, fixable waste
    condition, so the message can be sent straight to a facilities engineer.
    """
    findings = []
    empty = row["occupancy"] <= config.IDLE_OCCUPANCY

    if empty and row["hvac_kwh"] > config.HVAC_IDLE_LIMIT_KWH:
        findings.append(
            f"HVAC drawing {row['hvac_kwh']:.1f} kWh with {int(row['occupancy'])} "
            f"people present - suspected stuck damper or thermostat")

    if empty and row["machinery_kwh"] > config.MACHINERY_IDLE_LIMIT_KWH:
        findings.append(
            f"Machinery at {row['machinery_kwh']:.1f} kWh outside occupied hours "
            f"- equipment not shut down")

    if empty and row["lighting_kwh"] > config.LIGHTING_IDLE_LIMIT_KWH:
        findings.append(
            f"Lighting at {row['lighting_kwh']:.1f} kWh in an empty building "
            f"- lights left on")

    if row["total_kwh"] > config.CONTRACTED_DEMAND_KWH:
        findings.append(
            f"Demand {row['total_kwh']:.1f} kWh exceeds the contracted "
            f"{config.CONTRACTED_DEMAND_KWH:.0f} kWh - penalty applies")

    slab, _ = slab_for_hour(int(row["hour"]))
    if slab == "PEAK" and row["machinery_kwh"] > 12.0:
        findings.append(
            "Heavy machinery running inside the peak tariff window - "
            "consider shifting this batch to off-peak")

    return findings

def recommend(findings, deviation_pct):
    """Turn findings into the single most useful next action."""
    if not findings:
        if deviation_pct is not None and deviation_pct < -15:
            return "Consumption well below forecast - verify the meter is reporting"
        return "No action required"
    text = " ".join(findings).lower()
    if "hvac" in text:

```

```

        return "Dispatch technician to inspect the HVAC unit now"
    if "machinery" in text and "peak" in text:
        return "Reschedule the machinery batch to the off-peak window after 22:00"
    if "machinery" in text:
        return "Shut down idle machinery from the building management system"
    if "lighting" in text:
        return "Switch off unoccupied-zone lighting"
    if "demand" in text:
        return "Shed non-critical load to stay under the contracted demand"
    return "Review the flagged meter"

def price_dataframe(df):
    """Add all costing columns to a whole DataFrame at once."""
    priced = df.apply(cost_of_reading, axis=1, result_type="expand")
    out = df.join(priced)
    print(f"[tariff_engine] Priced {len(out)} readings -> "
          f"Rs.{out['total_cost'].sum():.2f} total, "
          f"{out['co2_kg'].sum():.1f} kg CO2")
    return out

if __name__ == "__main__":
    import data_loader

    demo = price_dataframe(data_loader.get_baseline())
    print(demo[["timestamp", "total_kwh", "tariff_slab", "tariff_rate",
                "total_cost", "co2_kg"]].head(8).to_string(index=False))

```

Element	Explanation
<code>slab_for_hour(hour)</code>	Walks the tariff table and returns the first slab whose hour set contains this hour. Membership in a <code>set</code> is <code>O(1)</code> , so the lookup cost does not grow if the utility adds more slabs.
<code>cost_of_reading(row)</code>	Builds the hourly bill from three independent parts — energy, fixed and penalty — and returns each separately as well as the total. Returning the parts is what lets the console print <i>'incl. Rs.306.00 demand penalty'</i> instead of one opaque number.
<code>max(0.0, kwh - CONTRACTED_DEMAND_KWH)</code>	The penalty applies only to the excess, never to the whole consumption. Getting this wrong is the most common mistake in demand billing code.
<code>check_rules(row)</code>	Returns a list of sentences, not a score. Each rule is a separate <code>if</code> , so adding a new fault condition never risks breaking an existing one. The <code>empty</code> flag is computed once and reused by three rules.
<code>recommend(findings, deviation_pct)</code>	Collapses possibly several findings into the single most useful next action, in priority order (HVAC fault first, then peak-window machinery, then idle machinery, then lighting, then load shedding). An operator who is handed five recommendations acts on none of them.
The negative-deviation branch	If consumption is far <i>below</i> forecast with no rule firing, the advice is to check the meter. An under-reporting sensor is as much a fault as an over-consuming machine, and a naive system only ever looks upward.
<code>price_dataframe(df)</code>	<code>df.apply(..., result_type="expand")</code> turns the dict returned per row into seven proper DataFrame columns in one pass.

10.4 eda.py — understanding the day before modelling

Never train on data you have not looked at. This module answers: where does the energy go, where does the *money* go (not the same question), and what actually drives the load?

SRC/EDA.PY

```

"""
eda.py
-----

```

Stage 3: describe the day in numbers, then in four pictures.

Uses matplotlib only, with the Agg backend so it runs on a server, in Docker and in Colab without trying to open a window.

```
"""

import matplotlib

matplotlib.use("Agg")
import matplotlib.pyplot as plt

import config

COLOURS = {
    "hvac_kwh": "#2b6cb0",
    "lighting_kwh": "#d69e2e",
    "machinery_kwh": "#c53030",
    "plug_load_kwh": "#2f855a",
}

SLAB_COLOUR = {"OFF-PEAK": "#2f855a", "NORMAL": "#d69e2e", "PEAK": "#c53030"}

def print_summary(df):
    print("\n" + "=" * 70)
    print("EXPLORATORY DATA ANALYSIS")
    print("=" * 70)

    total = df["total_kwh"].sum()
    peak_row = df.loc[df["total_kwh"].idxmax()]
    base_row = df.loc[df["total_kwh"].idxmin()]

    print(f"Readings           : {len(df)} hourly")
    print(f"Total consumption   : {total:.1f} kWh")
    print(f"Average per hour    : {df['total_kwh'].mean():.2f} kWh")
    print(f"Peak hour           : {peak_row['hour']:02d}:00 at "
          f"{peak_row['total_kwh']:.1f} kWh")
    print(f"Base load (min hour) : {base_row['hour']:02d}:00 at "
          f"{base_row['total_kwh']:.1f} kWh")
    print(f"Load factor         : {df['total_kwh'].mean() / peak_row['total_kwh']:.2f}"
          f" (average / peak; 1.0 would be perfectly flat)")
    print(f"Total cost          : Rs.{df['total_cost'].sum():.2f}")
    print(f"Total CO2           : {df['co2_kg'].sum():.1f} kg")

    print("\n-- Consumption by sub-meter --")
    for col in config.LOAD_COLUMNS:
        kwh = df[col].sum()
        print(f" {config.LOAD_LABELS[col]:<11} {kwh:7.1f} kWh   {kwh / total:6.1%}")

    print("\n-- Cost and energy by tariff slab --")
    by_slab = df.groupby("tariff_slab").agg(
        hours=("total_kwh", "size"),
        kwh=("total_kwh", "sum"),
        cost=("total_cost", "sum"),
    ).sort_values("cost", ascending=False)
    for slab, r in by_slab.iterrows():
        print(f" {slab:<9} {int(r['hours']):2d} h   {r['kwh']:6.1f} kWh   "
              f"Rs.{r['cost']:8.2f}   {r['cost'] / df['total_cost'].sum():5.1%} of bill")

    print("\n-- What drives the total (correlation) --")
    corr = df[["temperature_c", "occupancy", "is_working_hour",
               "total_kwh"]].corr(numeric_only=True)["total_kwh"].drop("total_kwh")
    for name, v in corr.sort_values(ascending=False).items():
        print(f" {name:<16} {v:+.3f}")

def _save(fig, name):
    path = config.OUTPUT_DIR / name
```

```

fig.tight_layout()
fig.savefig(path, dpi=130)
plt.close(fig)
print(f"[eda] saved {path.name}")

def build_charts(df):
    hours = df["hour"]

    # 1. Stacked load curve -- the single most important energy chart -----
    fig, ax = plt.subplots(figsize=(7, 3.6))
    bottom = 0
    for col in config.LOAD_COLUMNS:
        ax.fill_between(hours, bottom, bottom + df[col], label=config.LOAD_LABELS[col],
                        color=COLOURS[col], alpha=0.85, linewidth=0)
        bottom = bottom + df[col]
    ax.axhline(config.CONTRACTED_DEMAND_KWH, color="black", ls="--", lw=1.1,
               label=f"Contracted demand ({config.CONTRACTED_DEMAND_KWH:.0f} kWh)")
    ax.set_xlabel("Hour of day")
    ax.set_ylabel("kWh")
    ax.set_title("Daily load curve by sub-meter")
    ax.set_xticks(range(0, 24, 2))
    ax.legend(fontsize=8, loc="upper left")
    _save(fig, "01_load_curve.png")

    # 2. Cost per hour, coloured by tariff slab -----
    fig, ax = plt.subplots(figsize=(7, 3.2))
    ax.bar(hours, df["total_cost"],
           color=[SLAB_COLOUR[s] for s in df["tariff_slab"]])
    for slab, colour in SLAB_COLOUR.items():
        ax.bar(0, 0, color=colour, label=slab)
    ax.set_xlabel("Hour of day")
    ax.set_ylabel("Rupees")
    ax.set_title("Hourly cost, coloured by time-of-day tariff slab")
    ax.set_xticks(range(0, 24, 2))
    ax.legend(fontsize=8)
    _save(fig, "02_cost_by_slab.png")

    # 3. Temperature vs HVAC -----
    fig, ax = plt.subplots(figsize=(5, 3.6))
    sc = ax.scatter(df["temperature_c"], df["hvac_kwh"], c=df["occupancy"],
                   cmap="viridis", s=55, edgecolors="white")
    fig.colorbar(sc, ax=ax, label="Occupancy")
    ax.set_xlabel("Outside temperature (C)")
    ax.set_ylabel("HVAC consumption (kWh)")
    ax.set_title("HVAC load vs temperature")
    _save(fig, "03_hvac_vs_temperature.png")

    # 4. Share of consumption per sub-meter -----
    fig, ax = plt.subplots(figsize=(5, 3.6))
    totals = [df[c].sum() for c in config.LOAD_COLUMNS]
    labels = [config.LOAD_LABELS[c] for c in config.LOAD_COLUMNS]
    ax.barh(labels, totals, color=[COLOURS[c] for c in config.LOAD_COLUMNS])
    for i, v in enumerate(totals):
        ax.text(v + 0.6, i, f"{v:.1f} kWh ({v / sum(totals):.0%})",
               va="center", fontsize=8.5)
    ax.set_xlim(0, max(totals) * 1.35)
    ax.set_xlabel("kWh over the day")
    ax.set_title("Where the energy goes")
    _save(fig, "04_consumption_share.png")

def run(df):
    print_summary(df)
    build_charts(df)

```

```
if __name__ == "__main__":
    import data_loader
    import tariff_engine

    run(tariff_engine.price_dataframe(data_loader.get_baseline()))
```

Element	Explanation
<code>matplotlib.use("Agg")</code>	Selects a non-interactive backend before importing pyplot. Without it matplotlib tries to open a window and crashes on servers, in Docker and in CI.
Load factor	<code>mean / peak</code> . A value of 1.0 would mean a perfectly flat profile. Utilities love flat customers, and a low load factor is the clearest signal that demand management would pay.
Energy split vs cost split	The module deliberately prints both. They tell different stories: OFF-PEAK is a third of the hours but under 9% of the bill, while PEAK is a quarter of the hours and nearly 40%.
Stacked <code>fill_between</code>	The running <code>bottom</code> variable accumulates each sub-meter so the bands stack. The dashed line marks the contracted demand, turning an abstract number into something you can see the curve approach.
Conditional bar colours	<code>color=[SLAB_COLOUR[s] for s in df["tariff_slab"]]</code> colours each bar by its slab. The zero-height bars afterwards exist only to create legend entries — a standard matplotlib idiom.
Colour-mapped scatter	<code>c=df["occupancy"]</code> with a colour bar encodes a third variable, so one chart shows temperature, HVAC load and occupancy together.
<code>_save(fig, name)</code>	Private helper applying <code>tight_layout()</code> , 130 DPI and — crucially — <code>plt.close(fig)</code> . Forgetting to close figures inside a loop is the classic matplotlib memory leak.

10.5 model.py — forecasting and the anomaly band

The model's job is not to be clever. It is to give the anomaly detector a trustworthy expectation **and an honest measure of its own error**. The second half is what most tutorials skip.

SRC/MODEL.PY

```
"""
model.py
-----
Stage 4: learn what a normal hour looks like, so that the streaming layer can
recognise an abnormal one.

The model predicts total_kwh from the clock, the outside temperature and the
occupancy. Its job is not to be clever; its job is to give the anomaly detector
a trustworthy expectation and an honest error band.
"""

import joblib
import numpy as np
import pandas as pd
from sklearn.linear_model import LinearRegression
from sklearn.metrics import mean_absolute_error, r2_score
from sklearn.model_selection import KFold, cross_val_predict
from sklearn.pipeline import Pipeline
from sklearn.preprocessing import StandardScaler

import config

def build_pipeline():
    """
    Scale, then fit a straight-line model.

    StandardScaler matters because temperature (about 30), occupancy (about 70)
```

```

and the sine terms (between -1 and 1) live on wildly different scales.
Scaling puts them on equal footing so the coefficients can be compared.
"""
return Pipeline([
    ("scale", StandardScaler()),
    ("reg", LinearRegression()),
])

def train(df):
    """
    Evaluate with K-fold cross-validation, then refit on everything.

    Why not a plain train/test split? The baseline is a single 24-hour day.
    A chronological split would put the whole evening into the test set and the
    model would never have seen an evening hour. Cross-validation lets every
    hour take a turn as unseen data while still training on the rest.
    """
    X = df[config.FEATURES]
    y = df[config.TARGET]

    pipe = build_pipeline()
    cv = KFold(n_splits=config.CV_FOLDS, shuffle=True,
              random_state=config.RANDOM_STATE)

    # Out-of-fold predictions: every row predicted by a model that did not see it
    oof = cross_val_predict(pipe, X, y, cv=cv)
    residuals = y - oof

    metrics = {
        "rows": len(df),
        "folds": config.CV_FOLDS,
        "mae": round(float(mean_absolute_error(y, oof)), 3),
        "rmse": round(float(np.sqrt(np.mean(residuals ** 2))), 3),
        "r2": round(float(r2_score(y, oof)), 3),
        "mape": round(float(np.mean(np.abs(residuals / y)) * 100), 2),
        "sigma": round(float(residuals.std(ddof=1)), 3),
    }

    print("\n" + "=" * 70)
    print("FORECAST MODEL TRAINING")
    print("=" * 70)
    print(f"Rows / folds      : {metrics['rows']} / {metrics['folds']}")
    print(f"MAE                  : {metrics['mae']} kWh")
    print(f"RMSE                 : {metrics['rmse']} kWh")
    print(f"MAPE                 : {metrics['mape']} %")
    print(f"R-squared            : {metrics['r2']}")
    print(f"Residual sigma       : {metrics['sigma']} kWh <- the anomaly yardstick")
    print(f" WATCH band beyond : {config.WATCH_SIGMA * metrics['sigma']:.2f} kWh error")
    print(f" ALERT band beyond : {config.ALERT_SIGMA * metrics['sigma']:.2f} kWh error")

    # Refit on the complete baseline for deployment
    pipe.fit(X, y)
    joblib.dump({"pipeline": pipe, "sigma": metrics["sigma"], "metrics": metrics},
                config.MODEL_FILE)
    print(f"[model] saved to {config.MODEL_FILE.name}")

    return pipe, metrics

def coefficients(pipe):
    """Read the fitted weights back out; scaled inputs make them comparable."""
    coefs = pipe.named_steps["reg"].coef_
    table = (pd.DataFrame({"feature": config.FEATURES, "coefficient": coefs.round(3)})
             .assign(strength=lambda d: d["coefficient"].abs())
             .sort_values("strength", ascending=False)
             .drop(columns="strength"))

```

```

        .reset_index(drop=True))
print("\nWhat the model learned (kWh change per 1 standard deviation):")
print(table.to_string(index=False))
return table

def load():
    """Load the saved bundle, or explain how to create it."""
    if not config.MODEL_FILE.exists():
        raise FileNotFoundError("Model not trained yet. Run main.py first.")
    return joblib.load(config.MODEL_FILE)

def forecast_one(pipe, row):
    """Expected total_kwh for a single incoming reading."""
    frame = row.to_frame().T[config.FEATURES].astype(float)
    return float(pipe.predict(frame)[0])

def classify_deviation(actual, expected, sigma):
    """
    Compare what happened against what was expected, in sigma units.

    Using sigma rather than a fixed kWh threshold means the detector
    automatically becomes stricter when the model is accurate and more
    forgiving when it is not.
    """
    error = actual - expected
    z = error / sigma if sigma else 0.0
    if abs(z) >= config.ALERT_SIGMA:
        status = "ALERT"
    elif abs(z) >= config.WATCH_SIGMA:
        status = "WATCH"
    else:
        status = "NORMAL"
    pct = (error / expected * 100) if expected else 0.0
    return {
        "expected_kwh": round(expected, 2),
        "deviation_kwh": round(error, 2),
        "deviation_pct": round(pct, 1),
        "sigma_score": round(z, 2),
        "status": status,
    }

if __name__ == "__main__":
    import data_loader
    import tariff_engine

    frame = tariff_engine.price_dataframe(data_loader.get_baseline())
    model, _ = train(frame)
    coefficients(model)

```

Element	Explanation
Pipeline([scale, reg])	Bundles scaling and the estimator into one object, so the exact same transformation is applied at training and at prediction time. This single object is what gets saved to disk.
StandardScaler	Temperature is around 30, occupancy around 70, the sine terms between -1 and 1. Without scaling the coefficients would be incomparable and the model would be dominated by whichever feature happened to have the largest units.
KFold(shuffle=True, random_state=42)	Five folds over the 24 hours. Shuffling is safe here because the features already encode the time explicitly; the model is not using row order.

Element	Explanation
<code>cross_val_predict</code>	The most important line in the module. Every hour gets a prediction from a model that never saw that hour during fitting. The resulting residuals are therefore honest out-of-sample errors, not the artificially small in-sample ones.
Why not a train/test split	The baseline is a single 24-hour day. A chronological split would put the whole evening in the test set and the model would never have trained on an evening hour. Cross-validation lets every hour take a turn as unseen data while still training on the rest. <i>With several days of history the correct choice flips to a chronological split, or <code>TimeSeriesSplit</code>.</i>
<code>sigma = residuals.std(ddof=1)</code>	The spread of those out-of-fold errors becomes the yardstick for 'unusual'. It is computed from data the model did not see, which is what makes it trustworthy.
MAE, RMSE, MAPE, R-squared	Four views of the same error. MAE is the average miss in kWh, RMSE punishes big misses harder, MAPE expresses it as a percentage, and R-squared says how much of the variation the model explains.
<code>pipe.fit(X, y)</code> after evaluation	Evaluate on folds, then refit on everything. You want the deployed model to have seen all 24 hours; the fold scores told you how well it generalises, and that question is now answered.
<code>joblib.dump({pipeline, sigma, metrics})</code>	Saves a bundle , not just the model. The streaming layer needs sigma as much as it needs the pipeline; keeping them together means they can never drift apart.
<code>classify_deviation</code>	Divides the error by sigma to get a z-score, then maps it to NORMAL / WATCH / ALERT. Reporting the deviation in kWh, in per cent <i>and</i> in sigma covers three different audiences at once.
<code>forecast_one(pipe, row)</code>	<code>row.to_frame().T</code> converts a Series into a one-row DataFrame because scikit-learn always expects 2-D input; selecting <code>[config.FEATURES]</code> guarantees the column order matches training.

10.6 stream_simulator.py – the real-time engine

Everything before this was preparation. This module is what actually behaves like a live system.

SRC/STREAM_SIMULATOR.PY

```
"""
stream_simulator.py
-----
Stage 5: the real-time part.

A real building pushes a meter reading every few minutes over MQTT or Modbus.
We reproduce that by reading live_stream.csv one row at a time with a delay.
For every reading that arrives we:

    1. engineer exactly the same features used at training time
    2. price it through the tariff engine          -> money and carbon
    3. run the explainable operating rules         -> named waste conditions
    4. ask the model what it expected              -> deviation in sigma units
    5. combine rules and model into one status    -> NORMAL / WATCH / ALERT
    6. update rolling KPIs and append to the alert log

Only step 1's data source would change if this were fed by a real broker.
"""

import time
from datetime import datetime

import pandas as pd

import config
import data_loader
import model as model_module
import tariff_engine
```

```

def combine_status(model_status, findings):
    """
    Arbitration between the two layers.

    A named rule finding is hard evidence of a specific fault, so it can raise
    the severity on its own. The model can raise it too, but a purely
    statistical deviation with no rule behind it stays a WATCH until it is
    extreme, because the model is only as good as one day of history.
    """
    if findings and model_status == "ALERT":
        return "ALERT", "rule finding confirmed by the forecast model"
    if findings:
        return "ALERT", "operating rule breached"
    if model_status == "ALERT":
        return "WATCH", "large deviation from forecast, no rule matched"
    if model_status == "WATCH":
        return "WATCH", "mild deviation from forecast"
    return "NORMAL", "within the expected band"

class RollingKPI:
    """Running totals updated in O(1) per event, exactly like a stream job."""

    def __init__(self):
        self.readings = 0
        self.kwh = 0.0
        self.cost = 0.0
        self.co2 = 0.0
        self.penalty = 0.0
        self.alerts = 0
        self.watches = 0
        self.peak_kwh = 0.0
        self.peak_hour = None
        self.wasted_kwh = 0.0 # kWh above forecast on flagged hours

    def update(self, row, cost, status, deviation):
        self.readings += 1
        self.kwh += row["total_kwh"]
        self.cost += cost["total_cost"]
        self.co2 += cost["co2_kg"]
        self.penalty += cost["demand_penalty"]
        if status == "ALERT":
            self.alerts += 1
            if deviation["deviation_kwh"] > 0:
                self.wasted_kwh += deviation["deviation_kwh"]
        elif status == "WATCH":
            self.watches += 1
        if row["total_kwh"] > self.peak_kwh:
            self.peak_kwh = row["total_kwh"]
            self.peak_hour = int(row["hour"])

    def snapshot(self):
        return {
            "readings": self.readings,
            "kwh": round(self.kwh, 2),
            "cost": round(self.cost, 2),
            "co2": round(self.co2, 2),
            "penalty": round(self.penalty, 2),
            "alerts": self.alerts,
            "watches": self.watches,
            "peak_kwh": round(self.peak_kwh, 2),
            "peak_hour": self.peak_hour,
            "wasted_kwh": round(self.wasted_kwh, 2),
            "avg_rate": round(self.cost / self.kwh, 2) if self.kwh else 0.0,
        }

```

```

def run(bundle=None, delay=None):
    """Consume the whole stream and return (alert log, final KPI snapshot)."""
    bundle = bundle or model_module.load()
    pipe, sigma = bundle["pipeline"], bundle["sigma"]
    delay = config.STREAM_DELAY_SECONDS if delay is None else delay

    stream = data_loader.get_stream()
    kpi = RollingKPI()
    log = []

    print("\n" + "=" * 70)
    print("REAL-TIME METER STREAM")
    print("=" * 70)
    print(f"Model error band: 1 sigma = {sigma:.2f} kWh | "
          f"WATCH at {config.WATCH_SIGMA:.0f} sigma, ALERT at {config.ALERT_SIGMA:.0f} sigma")

    for _, row in stream.iterrows():
        time.sleep(delay)
        received = datetime.now().strftime("%H:%M:%S")

        cost = tariff_engine.cost_of_reading(row) # money
        findings = tariff_engine.check_rules(row) # rules
        expected = model_module.forecast_one(pipe, row) # model
        deviation = model_module.classify_deviation(
            row["total_kwh"], expected, sigma)
        status, why = combine_status(deviation["status"], findings)
        action = tariff_engine.recommend(findings, deviation["deviation_pct"])

        kpi.update(row, cost, status, deviation)
        snap = kpi.snapshot()

        mark = {"NORMAL": " ok ", "WATCH": " WATCH", "ALERT": " ALERT"}[status]
        print(f"\n[{received}] {row['timestamp']:%Y-%m-%d %H:%M} [{mark}]")
        print(f"    actual {row['total_kwh']:.1f} kWh | expected "
              f"{deviation['expected_kwh']:.1f} kWh | deviation "
              f"{deviation['deviation_kwh']:+.1f} kWh "
              f"{deviation['deviation_pct']:+.0f}%, {deviation['sigma_score']:+.1f} sigma)")
        print(f"    HVAC {row['hvac_kwh']:.1f} | Light {row['lighting_kwh']:.1f} | "
              f"Mach {row['machinery_kwh']:.1f} | Plug {row['plug_load_kwh']:.1f} | "
              f"{row['temperature_c']:.1f} C | {int(row['occupancy'])} people")
        print(f"    {cost['tariff_slab']:<8} @ Rs.{cost['tariff_rate']:.2f}/kWh -> "
              f"Rs.{cost['total_cost']:.2f}"
              + (f" (incl. Rs.{cost['demand_penalty']:.2f} demand penalty)"
                 if cost["demand_penalty"] else ""))
        for f in findings:
            print(f"    ! {f}")
        print(f"    verdict : {status} - {why}")
        print(f"    action : {action}")
        print(f"    LIVE KPI: {snap['readings']} readings | {snap['kwh']:.1f} kWh | "
              f"Rs.{snap['cost']:, .2f} | {snap['co2']:.1f} kg CO2 | "
              f"peak {snap['peak_kwh']:.1f} kWh at {snap['peak_hour']:02d}:00 | "
              f"{snap['alerts']} alerts")

    log.append({
        "received_at": received,
        "timestamp": row["timestamp"],
        "total_kwh": row["total_kwh"],
        "expected_kwh": deviation["expected_kwh"],
        "deviation_kwh": deviation["deviation_kwh"],
        "deviation_pct": deviation["deviation_pct"],
        "sigma_score": deviation["sigma_score"],
        "tariff_slab": cost["tariff_slab"],
        "total_cost": cost["total_cost"],
        "demand_penalty": cost["demand_penalty"],
        "co2_kg": cost["co2_kg"],
        "status": status,
    })

```

```

        "reason": why,
        "findings": " | ".join(findings) if findings else "",
        "action": action,
    })

log_df = pd.DataFrame(log)
log_df.to_csv(config.ALERT_LOG, index=False)
print(f"\n[stream] alert log written to {config.ALERT_LOG.name}")

final = kpi.snapshot()
wasted_cost = final["wasted_kwh"] * final["avg_rate"]
print("\n--- SESSION SUMMARY ---")
print(f"Readings processed      : {final['readings']}")
print(f"Energy consumed          : {final['kwh']:.1f} kWh")
print(f"Cost incurred             : Rs.{final['cost']:.2f} "
      f"(effective Rs.{final['avg_rate']:.2f}/kWh)")
print(f"Demand penalty          : Rs.{final['penalty']:.2f}")
print(f"Carbon emitted           : {final['co2']:.1f} kg CO2")
print(f"Peak demand              : {final['peak_kwh']:.1f} kWh at "
      f"{final['peak_hour']:02d}:00")
print(f"Alerts / watches         : {final['alerts']} / {final['watches']}")
print(f"Avoidable waste          : {final['wasted_kwh']:.1f} kWh "
      f"= about Rs.{wasted_cost:.2f} on flagged hours alone")

final["wasted_cost"] = round(wasted_cost, 2)
return log_df, final

if __name__ == "__main__":
    run()

```

Element	Explanation
<code>combine_status(model_status, findings)</code>	The arbitration policy of Section 9.5, in five lines. Note the deliberate asymmetry: a rule finding can raise the status on its own, but a purely statistical outlier is <i>downgraded</i> from ALERT to WATCH. With one day of history, the model has not earned the right to wake somebody up by itself.
<code>class RollingKPI</code>	Holds running totals rather than recomputing over the history. Each <code>update()</code> is $O(1)$, so cost per event stays constant no matter how long the stream runs — exactly how streaming aggregation works in Spark Structured Streaming or Flink.
<code>wasted_kwh</code>	Accumulates only the kWh <i>above forecast</i> on hours that were flagged. That is the defensible definition of avoidable waste: energy that was both unexpected and attributable to a named fault.
<code>snapshot()</code>	Returns the metrics as a plain dict, separating <i>state</i> from <i>presentation</i> . The console printer and the Streamlit dashboard both consume the same snapshot.
<code>time.sleep(delay)</code>	The only line that makes this a simulation rather than a batch job. - - <code>fast</code> sets it to zero; a real broker removes it entirely.
Four-step scoring per reading	Price, rule-check, forecast, arbitrate. If the model service were down, the tariff engine and the rules alone would keep the building monitored — a real resilience property, not a design flourish.
The printed block	Deliberately verbose: actual vs expected, the four sub-meter values, the tariff line, the findings, the verdict, the action, and the live KPIs. During a demonstration this is what makes the system look alive.
<code>log.append({...})</code> then <code>to_csv</code>	Builds the audit trail: timestamp, inputs, expectation, deviation, cost, verdict, findings and action for every single reading.

10.7 main.py — the orchestrator

Contains no analytics of its own. Its only job is to call the five stages in the right order, print readable banners, and write the summary. Keeping the orchestrator thin is what makes each stage independently testable.

SRC/MAIN.PY

```

"""
main.py
-----
Entry point. Runs the five stages in order:

    ingest -> tariff costing -> EDA -> forecast model -> real-time stream

Usage:
    python main.py          # full run, 1 second between meter readings
    python main.py --fast   # no delay
    python main.py --no-charts # console only, skip matplotlib
"""

import argparse
from datetime import datetime

import config
import data_loader
import eda
import model as model_module
import stream_simulator
import tariff_engine


def banner(step, title):
    print("\n" + "#" * 70)
    print(f"# STEP {step}: {title}")
    print("#" * 70)


def write_summary(priced, metrics, coefs, kpi):
    lines = [
        "SMART ENERGY ANALYTICS DASHBOARD - RUN SUMMARY",
        f"Generated : {datetime.now():%Y-%m-%d %H:%M:%S}",
        "",
        "1. BASELINE DAY",
        f"  Readings           : {len(priced)} hourly",
        f"  Consumption        : {priced['total_kwh'].sum():.1f} kWh",
        f"  Cost                : Rs.{priced['total_cost'].sum():.2f}",
        f"  CO2                 : {priced['co2_kg'].sum():.1f} kg",
        f"  Peak hour          : {int(priced.loc[priced['total_kwh'].idxmax(), 'hour']:02d):00}"
        f"  at {priced['total_kwh'].max():.1f} kWh",
        f"  Load factor        : {priced['total_kwh'].mean() / priced['total_kwh'].max():.2f}",
        "",
        "2. FORECAST MODEL",
        f"  MAE / RMSE          : {metrics['mae']} / {metrics['rmse']} kWh",
        f"  MAPE                : {metrics['mape']} %",
        f"  R-squared           : {metrics['r2']}",
        f"  Residual sigma      : {metrics['sigma']} kWh",
        "",
        "  Coefficients (per 1 standard deviation of the input):",
    ]
    for _, r in coefs.iterrows():
        lines.append(f"    {r['feature'][:<18]} {r['coefficient'][:>8]}")

    lines += [
        "",
        "3. REAL-TIME STREAM",
        f"  Readings processed : {kpi['readings']}",
        f"  Energy              : {kpi['kwh']:.1f} kWh",
        f"  Cost                : Rs.{kpi['cost']:.2f} (Rs.{kpi['avg_rate']:.2f}/kWh)",
        f"  Demand penalty      : Rs.{kpi['penalty']:.2f}",
        f"  CO2                 : {kpi['co2']:.1f} kg",
        f"  Peak                : {kpi['peak_kwh']:.1f} kWh at {kpi['peak_hour']:02d):00",
        f"  Alerts / watches    : {kpi['alerts']} / {kpi['watches']}",
    ]

```

```

        f"    Avoidable waste      : {kpi['wasted_kwh']:.1f} kWh "
        f"(about Rs.{kpi['wasted_cost']:.2f})",
        "",
        "Artefacts in outputs/: charts (*.png), load_forecast_model.joblib,",
        "realtime_alerts.csv, summary_report.txt",
    ]
    config.SUMMARY_REPORT.write_text("\n".join(lines), encoding="utf-8")
    print(f"\n[main] summary written to {config.SUMMARY_REPORT.name}")

def main():
    parser = argparse.ArgumentParser(description="Smart Energy Analytics Dashboard")
    parser.add_argument("--fast", action="store_true", help="no delay in the stream")
    parser.add_argument("--no-charts", action="store_true", help="skip PNG generation")
    args = parser.parse_args()

    print("\nSMART ENERGY ANALYTICS DASHBOARD")
    print(f"Started at {datetime.now():%Y-%m-%d %H:%M:%S}")

    banner(1, "DATA INGESTION & FEATURE ENGINEERING")
    baseline = data_loader.get_baseline()

    banner(2, "TARIFF COSTING & OPERATING RULES")
    priced = tariff_engine.price_dataframe(baseline)
    print(priced[["timestamp", "total_kwh", "tariff_slab", "tariff_rate",
                  "energy_cost", "total_cost", "co2_kg"]].to_string(index=False))

    banner(3, "EXPLORATORY DATA ANALYSIS")
    if args.no_charts:
        eda.print_summary(priced)
    else:
        eda.run(priced)

    banner(4, "LOAD FORECAST MODEL")
    pipe, metrics = model_module.train(priced)
    coefs = model_module.coefficients(pipe)

    banner(5, "REAL-TIME METER STREAM")
    bundle = {"pipeline": pipe, "sigma": metrics["sigma"], "metrics": metrics}
    _, kpi = stream_simulator.run(bundle, delay=0 if args.fast else None)

    write_summary(priced, metrics, coefs, kpi)
    print("\nPipeline finished successfully.\n")

if __name__ == "__main__":
    main()

```

Element	Explanation
<code>argparse</code>	Adds <code>--fast</code> and <code>--no-charts</code> , so the same script serves a live demonstration and a quick debug run without editing code.
<code>banner(step, title)</code>	A consistent header before each stage, so anyone watching your demo always knows which stage is running.
Passing the bundle to the stream	<code>main</code> hands the freshly trained pipeline and sigma straight to <code>stream_simulator.run()</code> instead of reloading from disk. The stream still works standalone, because <code>run()</code> falls back to <code>model.load()</code> when no bundle is given.
<code>write_summary(...)</code>	Persists the run to <code>summary_report.txt</code> . Console output scrolls away; a file can be attached to your submission.

Stage ordering

Element	Explanation
	Costing must precede EDA (the charts plot <code>total_cost</code> and <code>tariff_slab</code>) and must precede training (the priced frame carries every column forward). The stream runs last because it needs the fitted pipeline and its sigma.

11 How to Run the Project

11.1 Step by step

TERMINAL

```
# 1. unzip the project and open a terminal inside it
cd SmartEnergyAnalytics

# 2. (recommended) create an isolated environment
python -m venv venv
venv\Scripts\activate      # Windows
source venv/bin/activate   # macOS / Linux

# 3. install the five libraries
pip install -r requirements.txt

# 4. run the whole pipeline
cd src
python main.py
```

Command	Effect
<code>python main.py</code>	Full run with a 1-second pause between meter readings. Use this for your demonstration — the pauses make it visibly real-time.
<code>python main.py --fast</code>	Same output with no pauses. Finishes in about 3 seconds.
<code>python main.py --no-charts</code>	Skips matplotlib entirely. Useful on a machine where the plotting libraries misbehave.

11.2 Running a single module

TERMINAL

```
cd src
python data_loader.py      # ingestion + time features
python tariff_engine.py    # costing and operating rules
python eda.py              # regenerate the four charts
python model.py            # retrain and print coefficients
python stream_simulator.py # replay the live meter stream
```

11.3 Optional Streamlit dashboard

TERMINAL

```
pip install streamlit
python src/main.py      # must run once so the model exists
streamlit run app/dashboard.py
```

Three tabs: a **live meter stream** with animated KPI tiles and an actual-versus-expected line chart, a **baseline day** explorer with the load curve and hourly costs, and a **what-if calculator** where you set any hour's sub-meter values and see the bill, the penalty and the rule findings instantly. It imports the same `src/` modules, so no analytics logic is duplicated.

11.4 Running in Google Colab

COLAB NOTEBOOK CELL

```
from google.colab import files
files.upload()      # upload the zip
```

```
! unzip -q SmartEnergyAnalytics.zip
! cd SmartEnergyAnalytics/src
! pip install -q -r ../requirements.txt
! python main.py --fast
```

12 Expected Output

This is the real, unedited console output of `python main.py --fast`. If your run matches this, the project is working correctly. Only the timestamps and the `Started at` line will differ.

HEADER

```
SMART ENERGY ANALYTICS DASHBOARD
Started at 2026-07-28 01:16:20
```

STAGE 1 - INGESTION

```
# STEP 1: DATA INGESTION & FEATURE ENGINEERING
```

```
[data_loader] Loaded 24 readings from energy_readings.csv (2026-03-09 00:00:00 to 2026-03-09 23:00:00)
[data_loader] Validation -> {'readings': 24, 'duplicate_timestamps': 0, 'missing_values': 0, 'negative_readings': 0,
'interpolated': 0, 'interval_hours': 1.0, 'gaps_in_series': 0}
[data_loader] Engineered features; total load 326.2 kWh over 24 hours
```

STAGE 2 - TARIFF COSTING

```
# STEP 2: TARIFF COSTING & OPERATING RULES
```

```
[tariff_engine] Priced 24 readings -> Rs.3,095.86 total, 267.5 kg CO2
```

timestamp	total_kwh	tariff_slab	tariff_rate	energy_cost	total_cost	co2_kg
2026-03-09 00:00:00	4.0	OFF-PEAK	5.2	20.80	32.80	3.28
2026-03-09 01:00:00	3.9	OFF-PEAK	5.2	20.28	32.28	3.20
2026-03-09 02:00:00	3.8	OFF-PEAK	5.2	19.76	31.76	3.12
2026-03-09 03:00:00	3.8	OFF-PEAK	5.2	19.76	31.76	3.12
2026-03-09 04:00:00	3.9	OFF-PEAK	5.2	20.28	32.28	3.20
2026-03-09 05:00:00	4.5	OFF-PEAK	5.2	23.40	35.40	3.69
2026-03-09 06:00:00	6.5	NORMAL	7.5	48.75	60.75	5.33
2026-03-09 07:00:00	11.7	NORMAL	7.5	87.75	99.75	9.59
2026-03-09 08:00:00	17.9	NORMAL	7.5	134.25	146.25	14.68
2026-03-09 09:00:00	21.9	PEAK	10.8	236.52	248.52	17.96
2026-03-09 10:00:00	24.0	PEAK	10.8	259.20	271.20	19.68
2026-03-09 11:00:00	25.1	PEAK	10.8	271.08	287.58	20.58
2026-03-09 12:00:00	23.4	NORMAL	7.5	175.50	187.50	19.19
2026-03-09 13:00:00	25.2	NORMAL	7.5	189.00	210.00	20.66
2026-03-09 14:00:00	26.0	NORMAL	7.5	195.00	252.00	21.32
2026-03-09 15:00:00	25.7	NORMAL	7.5	192.75	236.25	21.07
2026-03-09 16:00:00	24.2	NORMAL	7.5	181.50	193.50	19.84
2026-03-09 17:00:00	21.2	NORMAL	7.5	159.00	171.00	17.38
2026-03-09 18:00:00	15.6	PEAK	10.8	168.48	180.48	12.79
2026-03-09 19:00:00	10.9	PEAK	10.8	117.72	129.72	8.94
2026-03-09 20:00:00	7.8	PEAK	10.8	84.24	96.24	6.40
2026-03-09 21:00:00	6.0	NORMAL	7.5	45.00	57.00	4.92
2026-03-09 22:00:00	4.9	OFF-PEAK	5.2	25.48	37.48	4.02
2026-03-09 23:00:00	4.3	OFF-PEAK	5.2	22.36	34.36	3.53

STAGE 3 - EXPLORATORY ANALYSIS

```
# STEP 3: EXPLORATORY DATA ANALYSIS
```

```
=====
EXPLORATORY DATA ANALYSIS
=====
```

```
Readings          : 24 hourly
Total consumption  : 326.2 kWh
Average per hour   : 13.59 kWh
Peak hour          : 14:00 at 26.0 kWh
Base load (min hour) : 02:00 at 3.8 kWh
Load factor        : 0.52 (average / peak; 1.0 would be perfectly flat)
Total cost         : Rs.3,095.86
Total CO2          : 267.5 kg
```

```
-- Consumption by sub-meter --
```

```

HVAC          112.4 kWh    34.5%
Lighting      56.7 kWh    17.4%
Machinery     105.7 kWh    32.4%
Plug load     51.4 kWh    15.8%

-- Cost and energy by tariff slab --
NORMAL    10 h    187.8 kWh    Rs.1,614.00    52.1% of bill
PEAK      6 h    105.3 kWh    Rs.1,213.74    39.2% of bill
OFF-PEAK  8 h    33.1 kWh    Rs. 268.12    8.7% of bill

-- What drives the total (correlation) --
occupancy      +0.991
temperature_c   +0.957
is_working_hour +0.946
[eda] saved 01_load_curve.png
[eda] saved 02_cost_by_slab.png
[eda] saved 03_hvac_vs_temperature.png
[eda] saved 04_consumption_share.png

```

STAGE 4 - FORECAST MODEL

STEP 4: LOAD FORECAST MODEL

```

=====
FORECAST MODEL TRAINING
=====
Rows / folds      : 24 / 5
MAE               : 0.384 kWh
RMSE              : 0.489 kWh
MAPE              : 4.91 %
R-squared         : 0.997
Residual sigma    : 0.497 kWh <- the anomaly yardstick
  WATCH band beyond : 0.99 kWh error
  ALERT band beyond : 1.49 kWh error
[model] saved to load_forecast_model.joblib

What the model learned (kWh change per 1 standard deviation):
  feature coefficient
  occupancy          6.015
  hour_cos           -2.281
  hour_sin           -1.288
  is_working_hour     0.644
  temperature_c      -0.169

```

STAGE 5 - REAL-TIME STREAM

STEP 5: REAL-TIME METER STREAM

```

[data_loader] Loaded 12 readings from live_stream.csv (2026-03-10 00:00:00 to 2026-03-10 11:00:00)
[data_loader] Validation -> {'readings': 12, 'duplicate_timestamps': 0, 'missing_values': 0, 'negative_readings': 0,
'interpolated': 0, 'interval_hours': 1.0, 'gaps_in_series': 0}
[data_loader] Engineered features; total load 150.2 kWh over 12 hours

=====
REAL-TIME METER STREAM
=====
Model error band: 1 sigma = 0.50 kWh | WATCH at 2 sigma, ALERT at 3 sigma

[01:16:21] 2026-03-10 00:00 [ ok ]
  actual    4.1 kWh | expected    3.9 kWh | deviation  +0.2 kWh (+5%, +0.4 sigma)
  HVAC 2.2 | Light 0.8 | Mach 0.5 | Plug 0.6 | 24.3 C | 0 people
  OFF-PEAK @ Rs.5.20/kWh -> Rs.33.32
  verdict : NORMAL - within the expected band
  action  : No action required
  LIVE KPI: 1 readings | 4.1 kWh | Rs.33.32 | 3.4 kg CO2 | peak 4.1 kWh at 00:00 | 0 alerts

[01:16:21] 2026-03-10 01:00 [ ok ]
  actual    4.0 kWh | expected    3.6 kWh | deviation  +0.4 kWh (+12%, +0.8 sigma)
  HVAC 2.1 | Light 0.8 | Mach 0.5 | Plug 0.6 | 23.9 C | 0 people
  OFF-PEAK @ Rs.5.20/kWh -> Rs.32.80
  verdict : NORMAL - within the expected band

```

```

action : No action required
LIVE KPI: 2 readings | 8.1 kWh | Rs.66.12 | 6.6 kg CO2 | peak 4.1 kWh at 00:00 | 0 alerts

[01:16:21] 2026-03-10 02:00 [ ok ]
actual 3.9 kWh | expected 3.5 kWh | deviation +0.4 kWh (+12%, +0.8 sigma)
HVAC 2.0 | Light 0.8 | Mach 0.5 | Plug 0.6 | 23.5 C | 0 people
OFF-PEAK @ Rs.5.20/kWh -> Rs.32.28
verdict : NORMAL - within the expected band
action : No action required
LIVE KPI: 3 readings | 12.0 kWh | Rs.98.40 | 9.8 kg CO2 | peak 4.1 kWh at 00:00 | 0 alerts

[01:16:21] 2026-03-10 03:00 [ ALERT]
actual 9.1 kWh | expected 3.6 kWh | deviation +5.5 kWh (+151%, +11.0 sigma)
HVAC 6.9 | Light 0.9 | Mach 0.6 | Plug 0.7 | 23.2 C | 0 people
OFF-PEAK @ Rs.5.20/kWh -> Rs.59.32
! HVAC drawing 6.9 kWh with 0 people present - suspected stuck damper or thermostat
verdict : ALERT - rule finding confirmed by the forecast model
action : Dispatch technician to inspect the HVAC unit now

... readings 05:00 to 10:00 follow the same format and were all NORMAL ...

actual 24.6 kWh | expected 24.4 kWh | deviation +0.2 kWh (+1%, +0.5 sigma)
HVAC 7.1 | Light 3.8 | Mach 9.8 | Plug 3.9 | 31.2 C | 69 people
PEAK @ Rs.10.80/kWh -> Rs.277.68
verdict : NORMAL - within the expected band
action : No action required
LIVE KPI: 11 readings | 118.4 kWh | Rs.1,096.67 | 97.1 kg CO2 | peak 24.6 kWh at 10:00 | 2 alerts

[01:16:22] 2026-03-10 11:00 [ ALERT]
actual 31.8 kWh | expected 25.7 kWh | deviation +6.1 kWh (+24%, +12.3 sigma)
HVAC 8.9 | Light 3.9 | Mach 14.6 | Plug 4.4 | 32.6 C | 72 people
PEAK @ Rs.10.80/kWh -> Rs.661.44 (incl. Rs.306.00 demand penalty)
! Demand 31.8 kWh exceeds the contracted 25 kWh - penalty applies
! Heavy machinery running inside the peak tariff window - consider shifting this batch to off-peak
verdict : ALERT - rule finding confirmed by the forecast model
action : Reschedule the machinery batch to the off-peak window after 22:00
LIVE KPI: 12 readings | 150.2 kWh | Rs.1,758.11 | 123.2 kg CO2 | peak 31.8 kWh at 11:00 | 3 alerts

[stream] alert log written to realtime_alerts.csv

--- SESSION SUMMARY ---
Readings processed : 12
Energy consumed : 150.2 kWh
Cost incurred : Rs.1,758.11 (effective Rs.11.71/kWh)
Demand penalty : Rs.306.00
Carbon emitted : 123.2 kg CO2
Peak demand : 31.8 kWh at 11:00
Alerts / watches : 3 / 0
Avoidable waste : 16.0 kWh = about Rs.187.13 on flagged hours alone

[main] summary written to summary_report.txt

Pipeline finished successfully.

```

12.1 Files produced

File in outputs/	Size	Contents
01_load_curve.png	~52 KB	Stacked daily load curve with the contracted-demand line
02_cost_by_slab.png	~28 KB	Hourly cost, each bar coloured by its tariff slab
03_hvac_vs_temperature.png	~38 KB	HVAC load against temperature, coloured by occupancy
04_consumption_share.png	~31 KB	Ranked kWh and percentage per sub-meter
load_forecast_model.joblib	~2 KB	The fitted pipeline, its sigma and its metrics
realtime_alerts.csv	~2.2 KB	One audit row per streamed reading
summary_report.txt	~1.1 KB	Plain-text KPIs for the whole run

13 Results and Interpretation

13.1 The baseline day at a glance

Metric	Value	What it tells you
Total consumption	326.2 kWh	One weekday for the whole building
Total cost	Rs. 3,095.86	An effective Rs. 9.49 per kWh once fixed charges are included
Carbon	267.5 kg CO ₂	Roughly what a small car emits over 1,500 km
Peak hour	14:00 at 26.0 kWh	Above the 25 kWh contract — the building is already paying penalties on its normal day
Base load	02:00 at 3.8 kWh	What the building draws with nobody in it. Every kWh of base load is paid for 8,760 hours a year.
Load factor	0.52	Average is only half of peak. A peaky profile like this is exactly where demand management pays.

13.2 What the charts show

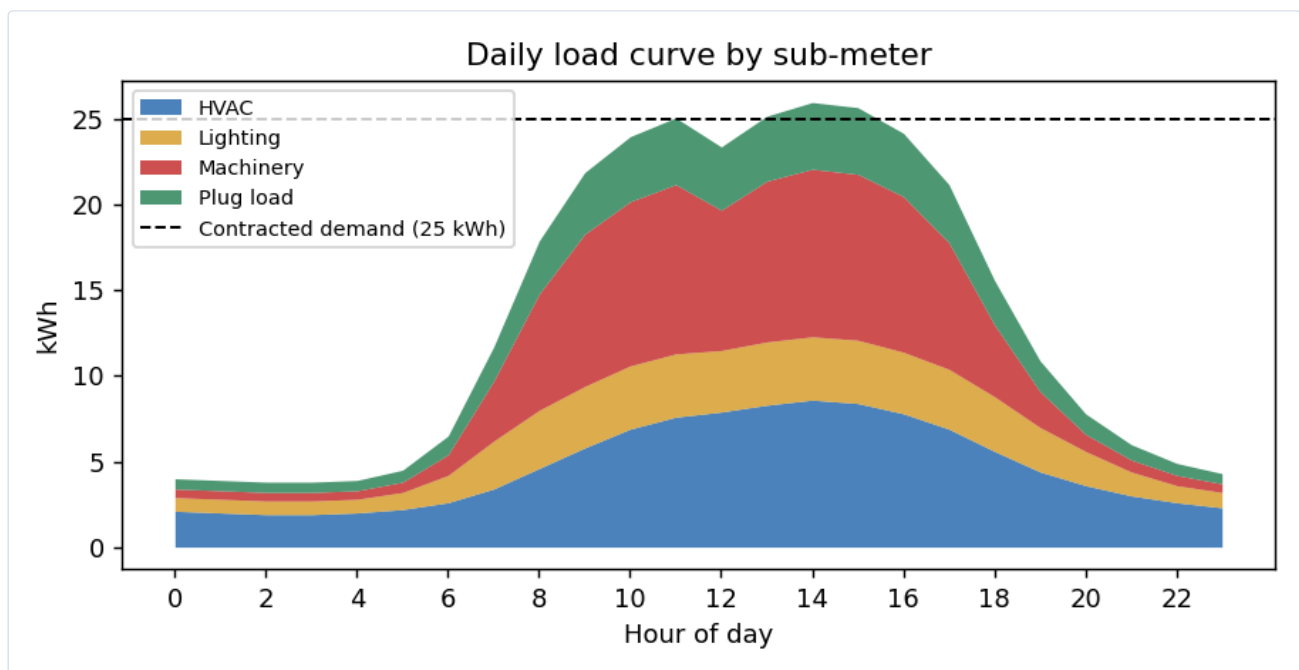


Fig 1 — Daily load curve by sub-meter, with the contracted-demand line

Figure 1. The classic office shape: a flat overnight base of about 4 kWh, a steep ramp from 06:00, a broad plateau through the working day, and a decline after 18:00. The dashed line is the 25 kWh contract, and the curve touches it between 13:00 and 15:00 — this building is running right at its limit on a perfectly ordinary day. HVAC (blue) grows steadily with the afternoon heat while machinery (red) tracks the shift pattern instead.

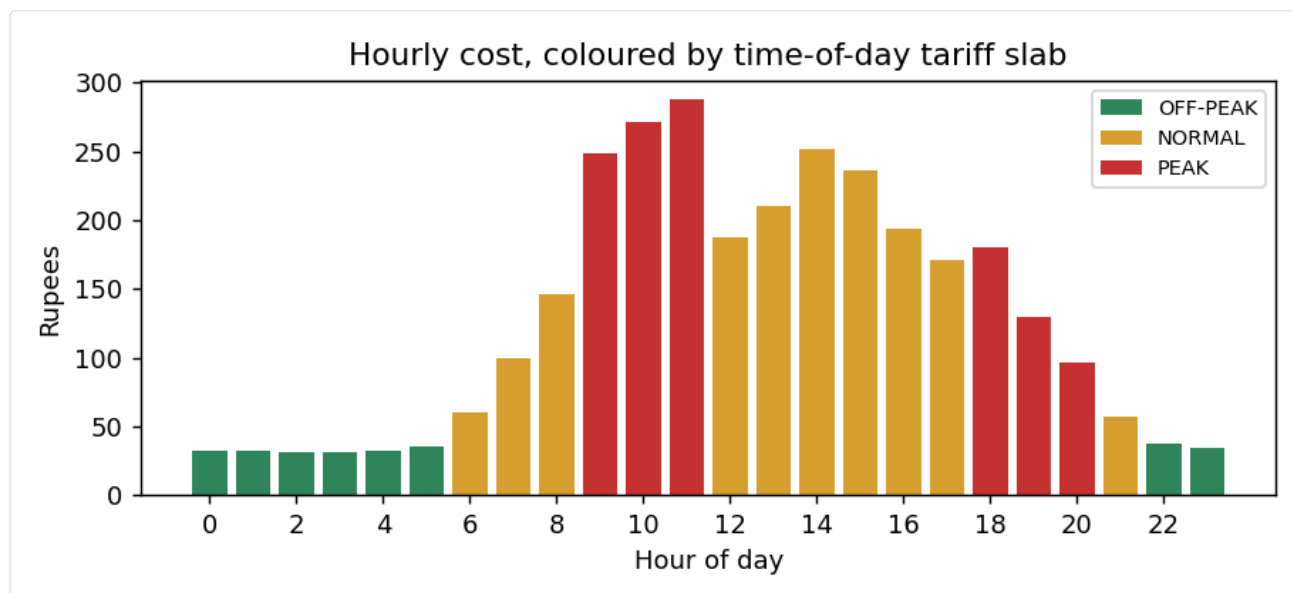


Fig 2 — Hourly cost coloured by tariff slab

Figure 2. Compare this with Figure 1 and the point of time-of-day pricing becomes obvious: the two shapes are *not the same*. The 10:00 and 11:00 bars are taller than 12:00 despite lower consumption, purely because they fall in the peak slab. Cost is driven by when you consume, not only how much.

Slab	Hours	Energy	Cost	Share of bill
NORMAL	10 h	187.8 kWh	Rs. 1,614.00	52.1%
PEAK	6 h	105.3 kWh	Rs. 1,213.74	39.2%
OFF-PEAK	8 h	33.1 kWh	Rs. 268.12	8.7%

The headline finding of the whole analysis

Six peak hours — a quarter of the day — carry **39% of the bill**, while the eight off-peak hours carry under 9%. Moving one hour of machinery from the peak window to overnight saves roughly Rs.5.60 on every kilowatt-hour shifted, with no reduction in output whatsoever. That is the recommendation the system makes automatically at 11:00 in the live stream.

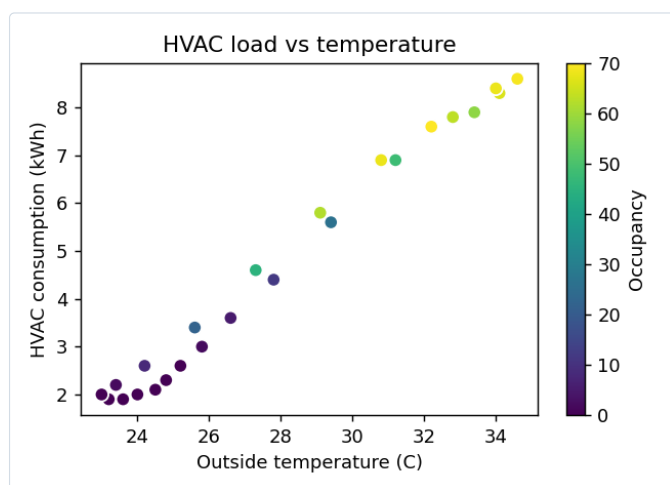


Fig 3 — HVAC vs temperature

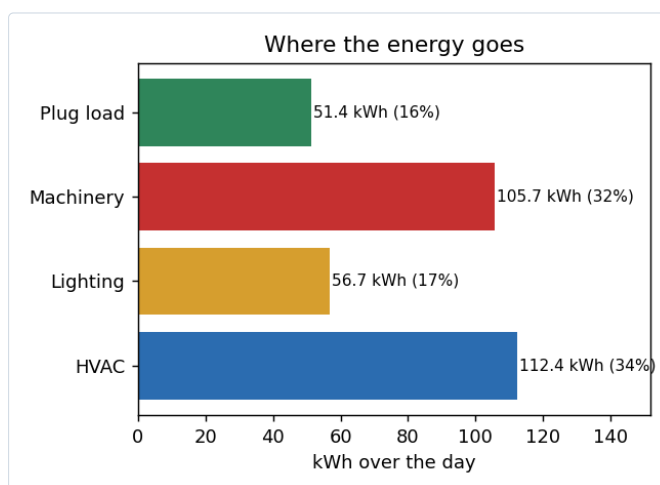


Fig 4 — Where the energy goes

Figure 3. An almost perfectly straight relationship: HVAC load rises with outside temperature, and the colour shows occupancy rising alongside it. That tight coupling is convenient physically and inconvenient statistically — see Section 13.4.

Figure 4. HVAC 34.5% and machinery 32.4% together account for two-thirds of consumption. An energy programme that starts with lighting (17.4%) is optimising the wrong third.

13.3 Forecast model performance

Metric	Value	How to read it
Rows / folds	24 / 5	Every hour predicted out-of-fold
MAE	0.384 kWh	The average miss, on hours averaging 13.6 kWh
RMSE	0.489 kWh	Close to the MAE, so there are no catastrophic individual misses
MAPE	4.91%	Under 5% error is considered good for hourly building load
R-squared	0.997	The model explains 99.7% of the variation in the load curve
Residual sigma	0.497 kWh	The anomaly yardstick. WATCH beyond 0.99 kWh of error, ALERT beyond 1.49 kWh

Why R-squared of 0.997 is not as impressive as it sounds

A single day's load curve is a smooth, highly predictable shape, and occupancy is effectively a proxy for it. The model has learned the shape of one day, not the behaviour of a building. It has never seen a weekend, a holiday, a monsoon day or a winter morning. Report this limitation explicitly — it is worth more marks than the number itself. Section 17 shows how to get a defensible estimate.

13.4 What the model learned — and a lesson in multicollinearity

Feature	Coefficient	Interpretation
occupancy	+6.015	By far the strongest driver. One standard deviation more people adds about 6 kWh to the hour.
hour_cos	-2.281	Captures the day/night swing: negative cosine means the middle of the day.
hour_sin	-1.288	Captures the morning/afternoon asymmetry.
is_working_hour	+0.644	The step change when the building switches on, over and above the smooth clock effect.
temperature_c	-0.169	Negative — and that should bother you. See below.

Temperature has a correlation of **+0.957** with total consumption, and Figure 3 shows an unmistakable positive relationship with HVAC load. Yet its regression coefficient is **negative**. This is not a bug; it is **multicollinearity**.

Temperature, occupancy and the time of day all rise and fall together in this dataset: the building is hottest exactly when it is fullest. Once the model has occupancy and the clock, temperature has almost no independent information left to contribute, so its coefficient is fitted to whatever tiny residual wobble remains — and the sign of that wobble is arbitrary.

What this means	What to do about it
The coefficient no longer answers 'what happens if temperature rises?'	To answer that, hold occupancy fixed — which this dataset never does
Individual coefficients are unstable, though predictions are fine	Judge the model by MAE and R-squared, not by the plausibility of one weight

What this means	What to do about it
The problem is the data, not the algorithm	Collect days where temperature and occupancy vary independently: weekends, holidays, a heatwave during a shutdown
It can be measured	Compute a variance inflation factor per feature, or drop occupancy and watch the temperature coefficient flip positive

Try this yourself

Remove "occupancy" from `config.FEATURES`, rerun, and watch the temperature coefficient become strongly positive while R-squared barely moves. Being able to explain that experiment is one of the most valuable things in this whole project.

13.5 Real-time session results

Twelve readings streamed. Three raised alerts; the pipeline consumed 150.2 kWh and Rs.1,758.11 in twelve hours — already more than half of a normal *full day's* bill.

Hour	Actual	Expected	Dev	Sigma	Slab	Cost Rs.	Status	Recommended action
00:00	4.1	3.92	+5%	+0.4	OFF-PEAK	33.32	NORMAL	-
01:00	4.0	3.58	+12%	+0.8	OFF-PEAK	32.80	NORMAL	-
02:00	3.9	3.48	+12%	+0.8	OFF-PEAK	32.28	NORMAL	-
03:00	9.1	3.62	+151%	+11.0	OFF-PEAK	59.32	ALERT	Dispatch technician to inspect the HVAC unit now
04:00	8.4	4.01	+110%	+8.8	OFF-PEAK	55.68	ALERT	Dispatch technician to inspect the HVAC unit now
05:00	4.6	5.0	-8%	-0.8	OFF-PEAK	35.92	NORMAL	-
06:00	6.8	7.19	-5%	-0.8	NORMAL	63.00	NORMAL	-
07:00	12.1	11.14	+9%	+1.9	NORMAL	102.75	NORMAL	-
08:00	18.4	18.1	+2%	+0.6	NORMAL	150.00	NORMAL	-
09:00	22.4	22.51	-0%	-0.2	PEAK	253.92	NORMAL	-
10:00	24.6	24.37	+1%	+0.5	PEAK	277.68	NORMAL	-
11:00	31.8	25.69	+24%	+12.3	PEAK	661.44	ALERT	Reschedule the machinery batch to the off-peak window after 22:00

KPI	Value	Comment
Energy	150.2 kWh	Twelve hours, midnight to noon
Cost	Rs. 1,758.11	An effective Rs. 11.71 per kWh, far above every published slab rate — the fixed charges and the penalty are the difference
Demand penalty	Rs. 306.00	All of it from the single 11:00 reading
Carbon	123.2 kg CO ₂	
Peak demand	31.8 kWh at 11:00	27% above the contract
Alerts / watches	3 / 0	03:00, 04:00 and 11:00
Avoidable waste	16.0 kWh ≈ Rs. 187	kWh above forecast on flagged hours only

13.6 The two incidents, read properly

Incident 1 — overnight HVAC fault (03:00 and 04:00). The building was empty and cool (23.2°C), so the model expected 3.6 kWh. The meter reported 9.1 kWh, of which 6.9 was HVAC alone. That is +151% and **+11.0 σ** . The rule engine independently named the cause, and the two layers agreeing produced a confident ALERT with a specific action. The fault persisted into 04:00 at +8.8 σ and then cleared. Total avoidable cost across the two hours: about Rs.52 of energy that heated nobody.

Incident 2 — peak-window machinery spike (11:00). Machinery jumped from 9.8 to 14.6 kWh, pushing the building total to 31.8 kWh against an expected 25.7 — **+12.3 σ** . Two rules fired at once: the demand breach and the peak-window machinery rule. The single hour cost Rs.661.44, of which **Rs.306.00 was pure penalty**. The recommended action is not 'use less' but 'run this batch after 22:00', which would move the same work to Rs.5.20 per kWh and remove the penalty entirely.

The difference between the two alerts

Incident 1 is *waste* — energy that produced nothing and should simply stop. Incident 2 is *mistimed useful work* — the energy is needed, just not at that hour. A system that reported both as 'anomaly' would be useless. Because the rule engine names the condition, the recommendation can differ: send a technician versus reschedule a batch.

14 Testing Checklist

Work down this list before you submit or demonstrate. Each row is something a reviewer might actually try.

#	Test	Expected result
T1	<code>python data_loader.py</code>	24 readings loaded, validation dict all zeros, interval 1.0 hour, 0 gaps
T2	Delete <code>data/energy_readings.csv</code> and run	<code>FileNotFoundError</code> naming the exact missing path
T3	Blank out one <code>hvac_kwh</code> cell and run	<code>interpolated: 1</code> , the value filled from its neighbours, the run continues
T4	Set one reading to a negative number	It is treated as a fault, blanked and interpolated — not summed into the total
T5	Delete the 12:00 row from the baseline file	<code>gaps_in_series</code> becomes non-zero, warning you the series is no longer continuous
T6	Shuffle the rows of the CSV and run	Identical results — the loader sorts chronologically
T7	Verify one hour's bill by hand from Section 9.2	03:00 = $9.1 \times 5.20 + 12.00$ = Rs. 59.32, matching the console
T8	Check <code>hour_sin</code> and <code>hour_cos</code> for hour 0 and hour 23	(0.0, 1.0) and (-0.259, 0.966) — adjacent on the circle
T9	<code>python tariff_engine.py</code>	24 readings priced, Rs.3,095.86 total, 267.5 kg CO ₂
T10	<code>python eda.py</code>	Four PNG files appear in outputs/
T11	<code>python model.py</code> twice	Identical MAE and sigma both times, proving <code>random_state</code> works
T12	Delete <code>outputs/load_forecast_model.joblib</code> then run <code>python stream_simulator.py</code>	A clear <code>FileNotFoundError</code> telling you to run <code>main.py</code> first
T13	Set <code>CONTRACTED_DEMAND_KWH = 20</code> and rerun	Many more demand breaches and a much larger penalty — the economics respond to configuration alone
T14	Set <code>ALERT_SIGMA = 6</code> and rerun the stream	The 03:00 and 11:00 alerts survive (11σ and 12σ); sensitivity is now demonstrably tunable
T15	Add a new row to <code>live_stream.csv</code>	It is scored automatically, with no code change
T16	Open <code>outputs/realtime_alerts.csv</code>	12 rows, each with a status, a reason and a recommended action

15 Common Errors and Fixes

Error message	Cause	Fix
ModuleNotFoundError: No module named 'config'	You ran the script from the project root instead of from inside src/	<code>cd src</code> first, then <code>python main.py</code>
ModuleNotFoundError: No module named 'sklearn'	Dependencies not installed, or installed into a different interpreter	<code>pip install -r requirements.txt</code> ; use <code>python -m pip</code> to be sure it is the same Python
FileNotFoundError: ...energy_readings.csv	The data folder was moved or renamed	Keep the folder structure from the zip intact
FileNotFoundError: Model not trained yet	You ran <code>stream_simulator.py</code> before ever training	Run <code>python main.py</code> once so the joblib file is created
AttributeError: Can only use .dt accessor with datetimelike values	The timestamp column was read as text	<code>parse_dates=["timestamp"]</code> is already in <code>load_csv</code> — make sure your edited CSV still has a readable timestamp format
ValueError: Cannot have number of splits <code>n_splits=5</code> greater than the number of samples	You cut the baseline file down to fewer than five rows	Keep at least ten readings, or lower <code>CV_FOLDS</code>
ValueError: Input X contains NaN	A column the interpolation does not cover was left blank	Only the numeric meter, temperature and occupancy columns are interpolated — do not blank the timestamp
Sigma comes out as 0 and every reading is ALERT	A perfectly fitting model on too few rows	Add more baseline hours; a zero error band makes every deviation infinite
Charts are blank or the script hangs	An interactive matplotlib backend is being used	<code>matplotlib.use("Agg")</code> is already set in <code>eda.py</code> — keep it above the pyplot import
TypeError: bar() got an unexpected keyword argument	matplotlib older than 3.7	Upgrade matplotlib, or check the argument name in your version
Rupee or degree symbols show as boxes in the console	Windows console code page	The code deliberately uses <code>Rs .</code> and <code>C</code> instead of the glyphs; keep it that way
Streamlit page loads but is empty	The model file does not exist yet	Run <code>python src/main.py</code> before launching the dashboard

16 Viva / Interview Questions with Answers

These are the questions most likely to be asked about this project. Read the answers, then close the handbook and try to say them in your own words.

Q1. What exactly is real-time about this project?

The processing model. Readings are handled one at a time as they arrive, each produces a decision immediately, and the aggregate KPIs update after every event rather than at the end of the file. The CSV plus a timer stands in for a message broker, but the event loop, the per-reading scoring path and the $O(1)$ rolling aggregation are the same design a production building management system uses.

Q2. Why do you have both a rule engine and a forecast model?

They fail in opposite ways. Rules are specific and explainable — they can say 'the HVAC is running in an empty building' — but they only catch faults somebody anticipated. The model is general: it flags any hour that does not look like the others, including faults nobody wrote a rule for. It just cannot say why. Running both gives named causes where rules exist and a safety net where they do not.

Q3. Why encode the hour as sine and cosine instead of a number from 0 to 23?

Because as plain numbers, 23:00 and 00:00 look 23 units apart when they are actually one hour apart. Mapping the clock onto a circle and handing the model the two coordinates makes midnight and 23:00 neighbours again. It is the reason the model reaches an R-squared of 0.997 on a curve that wraps around midnight, and the same trick applies to day of week, month, and compass bearing.

Q4. Why interpolate missing readings instead of filling with the median?

Because this is a time series with a strong shape. A missing 14:00 reading is best estimated from 13:00 and 15:00, both of which are near the peak. The day's median is around 13.6 kWh, which would badly under-fill a 26 kWh hour and badly over-fill a 4 kWh one. Interpolation respects the curve; a median flattens it.

Q5. Why cross-validation instead of a train/test split?

The baseline is a single 24-hour day. A chronological split would put the whole evening into the test set, so the model would never train on an evening hour and would be evaluated on a period it had no chance of learning. Cross-validation lets every hour take a turn as unseen data while still training on the rest. With several days of history the correct answer flips back to a chronological split or TimeSeriesSplit, because then you genuinely can hold out the future.

Q6. Where does sigma come from and why not just use a fixed kWh threshold?

Sigma is the standard deviation of the model's out-of-fold residuals — its honest error on data it did not see, 0.497 kWh in this run. Thresholds expressed in sigma adapt automatically: improve the model and the detector becomes more sensitive with no re-tuning. A hard-coded '2 kWh' silently becomes too loose or too tight the moment the model, the building or the season changes.

Q7. Your temperature coefficient is negative but temperature correlates +0.957 with load. Explain.

That is multicollinearity. Temperature, occupancy and the time of day all move together in this dataset — the building is hottest exactly when it is fullest. Once occupancy and the clock are in the model, temperature has almost no independent information left, so its coefficient is fitted to a tiny residual wobble whose sign is essentially arbitrary. The predictions are still fine; it is the individual weight that is unstable. Dropping occupancy makes the temperature coefficient flip strongly positive.

Q8. Why is a demand penalty charged only on the excess?

Because the contract is a commitment to stay below a level, not a price band. Exceeding 25 kWh by 6.8 kWh costs 6.8 times the penalty rate, not 31.8 times. Applying the penalty to the whole consumption is the single most common bug in demand-billing code, and it inflates the number by roughly five times here.

Q9. Why does the effective rate come out at Rs.11.71 per kWh when the highest published slab is Rs. 10.80?

Two reasons. The fixed charge of Rs.12 per hour is levied regardless of consumption, so it inflates the per-kWh figure heavily during low hours. And the Rs.306 demand penalty at 11:00 is spread across the whole session. This is why energy managers watch the effective rate, not the tariff card — it is the only number that reflects what they actually pay.

Q10. What is load factor and why does 0.52 matter?

Load factor is average demand divided by peak demand. At 0.52 the building averages only half of what it peaks at, which means the infrastructure and the demand contract are both sized for a level reached in only a few hours a day. Flattening the profile — shifting machinery overnight — lets the contract be renegotiated downward, which saves money even before a single kilowatt-hour is saved.

Q11. Why does a purely statistical ALERT get downgraded to WATCH?

Because the model has one day of history. It has never seen a weekend, a holiday or a season, so an unusual-looking hour may simply be an hour it has no experience of. A named rule finding is hard evidence and can raise severity alone; a bare statistical outlier is a hint. With a year of data you would remove that downgrade.

Q12. How would you deploy this?

Replace the CSV reader with an MQTT or Modbus consumer, wrap the scoring path in a FastAPI endpoint, containerise with Docker, write readings and verdicts to a time-series database such as InfluxDB or TimescaleDB instead of CSV, push the rolling KPIs to Prometheus and visualise in Grafana, and route ALERT events to a paging system. The src/ modules themselves would barely change, which is the point of the layered design.

Q13. How would you detect that the forecast model has gone stale?

Monitor the residuals continuously. If the rolling mean of the error drifts away from zero, the building's behaviour has changed — new equipment, a new shift pattern, a different season. If the rolling standard deviation grows, the model is losing precision and sigma needs recomputing. Retrain on a schedule and whenever either statistic crosses a threshold.

Q14. What are the limitations of this project as it stands?

One day of baseline data, one building, one season, no weekends and no holidays. The model has learned the shape of a single weekday. Occupancy is assumed to be measured accurately, when in practice it is often inferred. The tariff is simplified — no taxes, duties or power-factor adjustments. And the anomaly detector can only compare against what it has seen, so a fault that was present throughout the baseline day would be learned as normal.

Q15. How would you extend this to a whole campus?

Add a building or meter identifier to every reading and group by it, train one model per building or a single model with the building as a categorical feature, aggregate KPIs at both building and campus level, and rank buildings by kWh per square metre so the worst performer is obvious. The streaming loop stays the same; only the grouping changes.

17 Extension Ideas

Ordered from a single evening's work to a full second project. Pick two or three and you have a genuinely distinctive submission.

Level	Extension	What it teaches
Easy	Drop <code>occupancy</code> from the features and watch the temperature coefficient flip positive	Multicollinearity, demonstrated rather than described
Easy	Generate a second and third baseline day (a weekend, a hot day) and retrain	How sample size and diversity change a model
Easy	Add a fifth chart: actual versus expected kWh across the stream, with the $\pm 2\sigma$ band shaded	Visualising an anomaly detector
Easy	Add a solar-generation column and subtract it from the billable total	Net metering arithmetic
Medium	Compute the exact saving from shifting the 11:00 machinery batch to 23:00, including the avoided penalty	Turning an alert into a costed business case
Medium	Replace <code>KFold</code> with <code>TimeSeriesSplit</code> once you have several days	Why time-series validation must not shuffle
Medium	Add a second model (RandomForest or gradient boosting) and compare MAE	Model selection and the bias/variance trade-off
Medium	Log every reading into SQLite with a proper schema instead of CSV	Persistence and basic data engineering
Medium	Expose the scoring path through a FastAPI <code>/reading</code> endpoint and post readings with curl	Turning analytics into a service
Medium	Add an email or webhook notifier that fires only on ALERT	Alert routing and avoiding notification fatigue
Hard	Implement drift detection on the residual mean and standard deviation, with automatic retraining	Production ML monitoring
Hard	Add a day-ahead forecast and a simple load-shifting optimiser that proposes a schedule minimising cost under the demand cap	Optimisation on top of prediction
Hard	Replace the CSV reader with a real MQTT broker running in Docker and publish readings from a second script	Genuine stream processing infrastructure
Hard	Extend to a campus: multiple buildings, per-building models, and a kWh-per-square-metre league table	Multi-entity analytics and benchmarking

18 Evaluation Rubric

A suggested 100-mark scheme. Useful for instructors, and useful for students as a self-check before submitting.

Criterion	Marks	What full marks looks like
Setup and execution	10	The project runs end to end from a clean environment using only the instructions in the README
Time-series data handling	12	Timestamp parsing, chronological sorting, interval detection and interpolation all explained, with a clear reason why interpolation beats a median here
Feature engineering	13	The cyclical sine/cosine encoding explained properly, with the hour-23 versus hour-0 distance worked through
Tariff and rules engine	15	One hourly bill reproduced by hand including the penalty arithmetic; each operating rule justified
Exploratory analysis	10	All four charts interpreted, and the energy-split versus cost-split distinction articulated
Forecast model	15	Pipeline, scaling, cross-validation and every metric explained; the origin of sigma understood
Real-time implementation	15	The event loop, the arbitration policy and the $O(1)$ rolling KPIs explained, and the demo visibly processes readings one at a time
Critical thinking	10	The one-day limitation acknowledged, the negative temperature coefficient explained as multicollinearity, and at least one extension attempted

Where marks are most often lost

Not in the code — the code is given to you. Marks are lost when a student cannot explain *why* a design choice was made: why sine and cosine, why cross-validation instead of a split, why sigma instead of a fixed threshold, why a rule finding can overrule the model. Section 16 covers every one of those questions.

19 Glossary

Term	Meaning
kWh	Kilowatt-hour: one kilowatt drawn for one hour. The unit electricity is sold in.
Sub-meter	A meter measuring one system (HVAC, lighting) rather than the whole building, so waste can be attributed.
Base load	The consumption that never goes away — what the building draws with nobody in it.
Peak demand	The highest demand reached in a billing period. Often priced separately from total consumption.
Contracted demand	The maximum demand a customer commits to. Exceeding it triggers a penalty on the excess.
Load factor	Average demand divided by peak demand. Closer to 1.0 means a flatter, cheaper-to-serve profile.
Load curve	Consumption plotted against time of day; the fingerprint of how a building is used.
Time-of-day (TOD) tariff	A rate that changes by hour, higher when the grid is stressed.
Emission factor	Kilograms of CO ₂ released per kWh generated, specific to a grid.
Cyclical encoding	Representing a repeating quantity such as the hour as sine and cosine so its ends join up.
Interpolation	Estimating a missing value from its neighbours in the series rather than from a summary statistic.
Pipeline (scikit-learn)	An object chaining preprocessing and an estimator so both are fitted and applied together.
Standardisation	Rescaling a numeric column to mean 0 and standard deviation 1 so features on different scales become comparable.
Cross-validation	Splitting data into folds so every row is predicted once by a model that did not train on it.
Out-of-fold residual	The error on a row predicted by a model that never saw it — an honest estimate of real-world error.
MAE / RMSE / MAPE	Mean absolute error, root mean squared error, mean absolute percentage error. Three ways to summarise miss size.
R-squared	The share of the variation in the target that the model explains; 1.0 is perfect.
Sigma (σ)	Standard deviation. Here, the spread of the model's own errors, used as the anomaly yardstick.
Multicollinearity	When input features move together, making individual coefficients unstable even though predictions stay accurate.
Rolling aggregate	A statistic updated incrementally as each event arrives, rather than recomputed from scratch.
Straight-through processing	An event handled entirely by software with no human intervention.
Audit trail	A permanent, timestamped record of every automated decision and the reasoning behind it.

End of handbook — Smart Energy Analytics Dashboard.

Everything described here is included in the accompanying zip file. Run it, break it, extend it.